SECP2523: DATABASE

SEM I 2025/ 2026

DATABASE DESIGN DESCRIPTIONS: LOGICAL DATABASE DESIGN

FOR HASTA CAR RENTAL SYSTEM

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To ensure legibility, high-resolution versions of all figures and diagrams included in this report can be accessed via this Google Drive link: High-Resolution Report Photos

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1.0 Introduction

This phase focuses on the database logical design of the Hasta Car Rental System based on the requirements and conceptual design established in Phase 2. The objective of this phase is to translate the conceptual Entity-Relationship Diagram (ERD) into a logical database structure that can be directly implemented using a relational database management system.

The logical design process involves refining the conceptual model by resolving non-relational features such as many-to-many relationships, identifying primary and foreign keys, and transforming entities into normalized relational tables. Normalization is performed to eliminate data redundancy and improve data consistency up to Third Normal Form (3NF). The relations are further evaluated against Boyce-Codd Normal Form (BCNF), and no violations are identified where applicable.

Additionally, this phase derives relational database schemas, updates the data dictionary according to the normalized tables as well as validates the logical design against the system's transaction requirements. SQL Data Definition Language (DDL) and Data Manipulation Language (DML) statements are also proposed to demonstrate how the logical design supports the required system operations.

Overall, this phase provides a structured and implementable database design that serves as a foundation for database implementation and system development in the subsequent phase.

2.0 Overview of project

The Hasta Car Rental System is a web-based fleet management system developed for Hasta Travel & Tours (Sdn. Bhd.), operating within Universiti Teknologi Malaysia (UTM). The system is designed to digitalize the company's current manual operations, which rely on disjointed tools like WhatsApp, physical whiteboards, and Excel spreadsheets. By implementing a centralized Relational Database Management System (RDBMS) using MySQL, the project aims to streamline workflows for students, staff, and top management, ensuring real-time data accuracy and operational efficiency.

The system facilitates the entire rental lifecycle—from secure user registration and real-time vehicle booking to payment verification, digital vehicle inspection, and deposit refunds. It enforces critical business rules, such as mandatory document verification (license, matric ID), automated penalty logic for fuel shortages or damages, and a loyalty program that rewards frequent renters.

2.1 System Scope and Core Modules

The system is structured into five modules to support operational requirements:

- User Management Module: Handles registration and authentication using UTM Matric/Staff IDs. It incorporates a Blacklist System to restrict access for users with outstanding violations and a Loyalty System that tracks rental history, automatically awarding stamps for eligible bookings.

- Vehicle Management Module: Manages the fleet inventory by storing vehicle information such as vehicle specifications, maintenance logs, and document validity (Road Tax/Insurance). It features an automated availability check that updates vehicle status in real-time to prevent double-booking.

- Booking Module: Enables customers to search for vehicles, view real-time availability, and place bookings with a minimum 1-hour duration. It manages the full booking lifecycle, including digital Pickup and Return Inspection Forms where users must upload photos of the vehicle condition and fuel levels.

- Payment Module: Facilitates financial transactions via DuitNow QR with manual receipt verification. It tracks rental fees, security deposits, and fines, ensuring a clear financial audit trail for every transaction.

- Reporting Module: Provides data visualization for Top Management. This includes Financial Reports (revenue, expenses, and net profit) and Demographic Insights, which categorize bookings by Faculty and College to identify key customer groups.

2.2 User Roles and Access Control

The system implements Role-Based Access Control (RBAC) to ensure data security and operational integrity:

1. Customer: Can register, upload verification documents (Matric Card, IC, License), view vehicle availability, make bookings, track loyalty stamps, and submit inspection forms.
2. Staff (Admin): Responsible for daily operations, including verifying customer documents and payments, approving bookings, managing fleet maintenance, and issuing penalties for damages or late returns.
3. Top Management: Has exclusive access to view sensitive financial reports, approve staff reimbursement claims, and oversee overall business performance and payroll data.

3.0 Database conceptual design

3.1 Updated business rule

Business Rules	Description
Customer Eligibility & Verification	**UTM Exclusivity**: Only UTM members (students or staff) are allowed to register. A valid Matric Number or Staff ID is required. **Document Verification**: A customer cannot be registered unless they upload IC/Passport, Driving License, and Matric ID/Staff ID.
Booking & Reservation Constraints	**Advance Notice**: Bookings must be made at least **12 hours** in advance. **Minimum Duration**: The rental duration must be at least **1 hour**. **Availability Integrity**: Vehicles cannot be booked if they are already **Approved**, **Rented**, or **Under Maintenance** during the selected dates.
Payment & Deposit Logic	**Tiered Deposit:** Rentals less than 15 days: RM 50 deposit. Rentals ≥ 15 days: The deposit is 100% of the rental fee. **Proof-of-Payment:** A booking becomes "Active/Waiting Verification" only after uploading a valid receipt.
Return & Penalty Calculations	**Manual Penalty Issuance:** Penalties for late returns are applied manually by staff, based on the specific circumstances.
Loyalty Program Automation	**Stamp Accumulation:** 1 stamp is awarded for every 3-hour or more rental. Reward Milestones: Vouchers are automatically issued at 3, 6, 9 stamps milestones, with a final reward after 15 stamps.

3.2 Conceptual ERD

According to the data modeling guidelines outlined by Visual Paradigm Online (n.d.), a conceptual data model focuses on identifying key entities and relationships at a high level, without including detailed attributes.

Based on this approach, the conceptual model in this project intentionally excludes attributes and serves to illustrate the overall structure of the system. Attribute details, keys, and constraints are subsequently defined in the logical and physical data models to maintain clarity and proper separation of concerns.

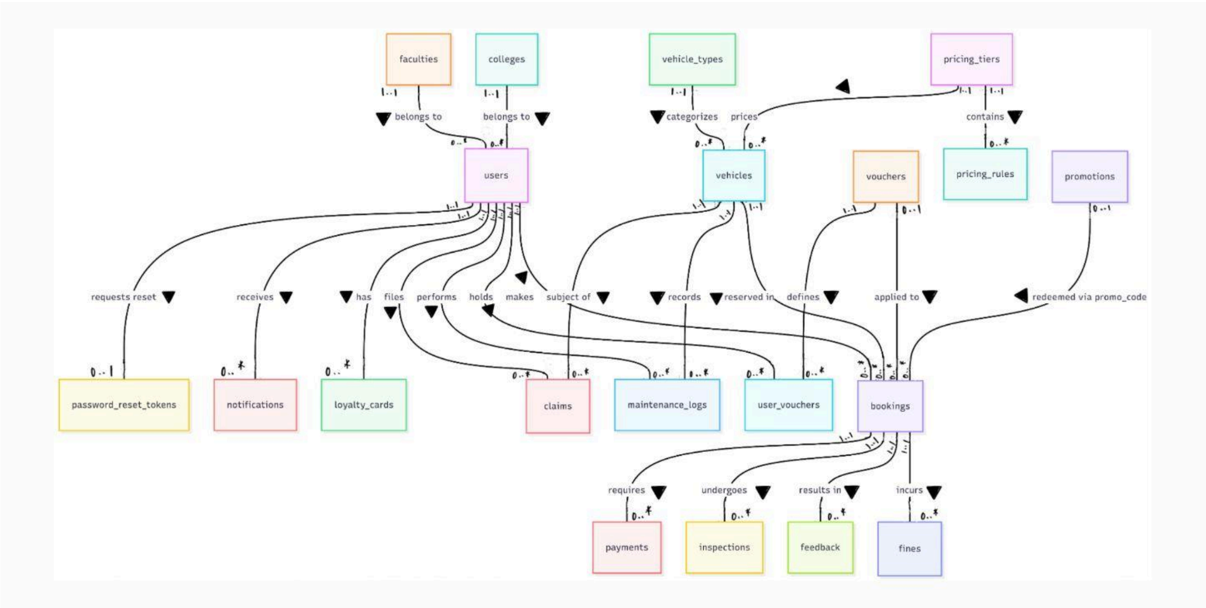

Figure 3.2.1: Conceptual ERD

4.0 DB logical design

This section details the logical design phase of the database development for the Hasta Travel & Tours system. While the conceptual design in Section 3.0 focused on high-level business rules and entities, this section translates those requirements into a structured logical model. It defines the specific attributes, primary keys, foreign keys, and relationships required to support the system's data integrity. Additionally, this chapter covers the data dictionary and the normalization process.

4.1 Logical ERD

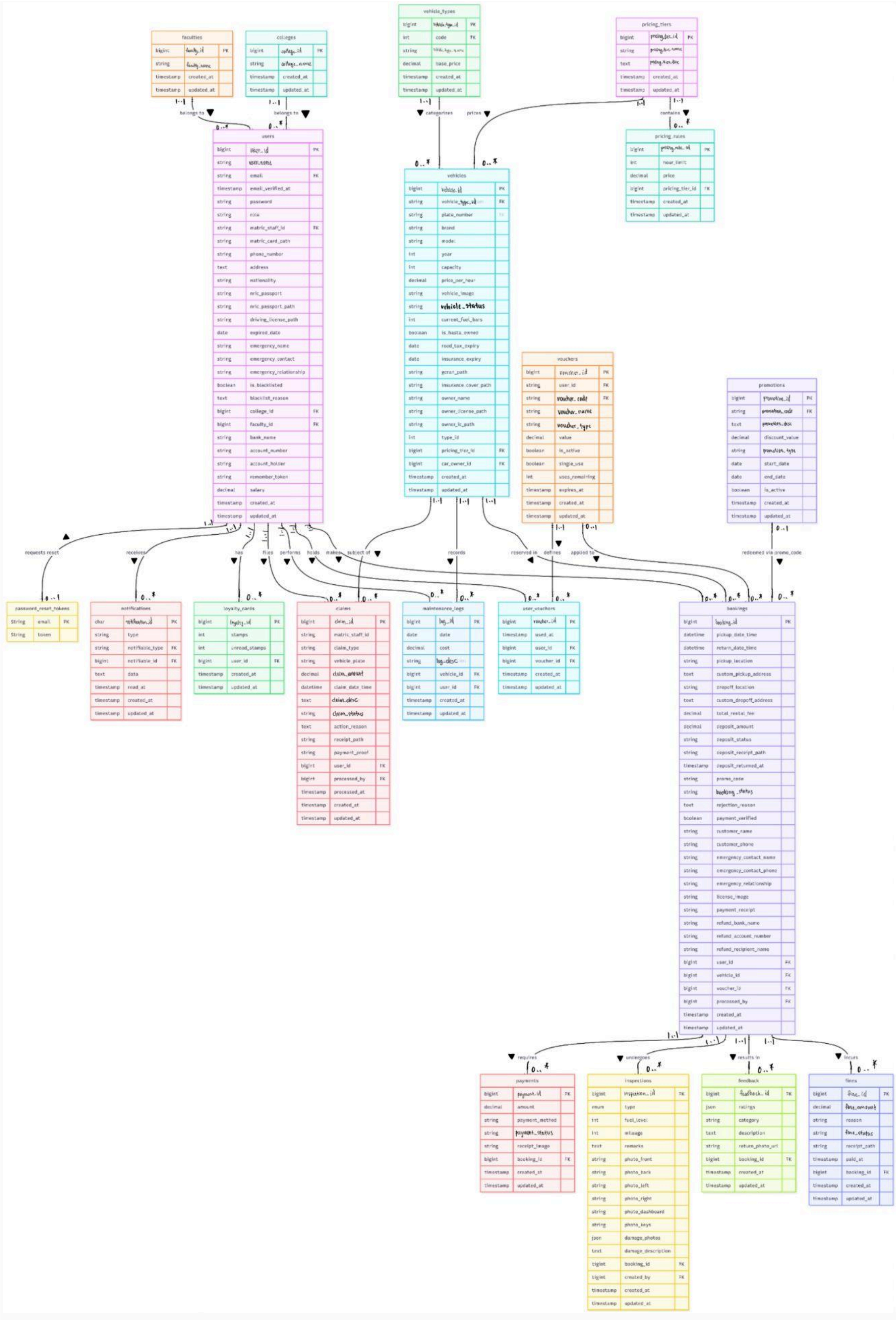

Figure 4.1.1: Logical ERD

4.2 Updated Data Dictionary

4.2.1 Data Description

Database is significant and can be identified as the basis of a system to store and support the smooth and well integrated system process. It consists of lots of tables that include information domains and represent the major entities. Each table has its primary key as their unique. The relationship between entities is maintained by using foreign keys. This ensures data integrity and supports the processing logic of the system. In our project, the major data or systems entities are stored into a relational database named carrentalsystem, processed and organized into 26 entities as listed in Table 4.2.1.1.

Table 4.2.1.1: Description of Entities in the Database

No.	Entity Name	Description
1	bookings	Records rental details, including pickup and return schedules, payment proofs and the approval status
2	claims	Records the financial claims for the admin staff for delivery tasks,fuel, or other reimbursement
3	colleges	Records customers with their corresponding colleges so that can be collected as the data to visualize at the admin or top manager dashboard
4	faculties	Records customers with their corresponding faculties so that can be collected as the data to visualize at the admin or top manager dashboard
5	feedback	Stores customers feedbacks, ratings and review regarding vehicle condition after a booking
6	fines	Records penalties issued to specific bookings, including the fine amount, reason, evidence receipts, and payment status
7	inspections	Stores vehicle inspection records (checklist and photos) conducted before and after a rental to track vehicle condition
8	loyalty_cards	Tracks user loyalty participation by storing the count of accumulated stamps earned

		based on business rules which is receiving 1 stamp for rentals exceeding 3 hours for reward redemption
9	maintenance_logs	Records the history of vehicle maintenance, including dates, costs, as well as descriptions of service or repairs
10	notifications	Stores system notifications sent to users, including the message content and read status
11	password_reset_tokens	Stores the temporary OTP codes generated for password recovery, allowing the system to verify user identity and enforce expiration limits 15 minutes
12	payments	Records transaction details including amount, payment method, uploaded receipt images and verification status
13	pricing_rules	Stores the price list for specific time blocks (like 1 hour,12 hours, 24 hours) for every pricing tier
14	pricing_tiers	Groups vehicles into tier categories so that prices can be managed collectively
15	promotions	Stores details of stamp collected and their corresponding rewards and display on the system to attract users
16	users	Stores account information for all system users, including login credentials and profile data.
17	user_vouchers	A bridge table that links specific vouchers to the users who have claimed or used them
18	vehicles	Stores detailed information about the cars available for rent, including plate number, model, color, price rate, and availability status
19	vehicle_types	Categorizes vehicles into types such as Sedan, Compact and Hatchback to provide the filter features for customers
20	vouchers	Defines redeemable vouchers with specific codes, values and usage limits for users

4.2.2 Data Dictionary

4.2.2.1 Entity: bookings

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for each booking transaction.
user_id	bigint	Foreign key. References the users table to identify the account holder who made the reservation
vehicle_id	bigint	Foreign key. References the vehicles table to identify the specific car being rented
pickup_date_time	datetime	Scheduled start date and time of the rental. Used to calculate the total rental duration.
return_date_time	datetime	Scheduled end date and time. Must be strictly later than the pick_up_date time.
pickup_location	varchar(50)	General location name for pickup
custom_pickup_address	text	Specific address details provided by the user when pickup location is not Student Mall as default
dropoff_location	varchar(50)	General location name for dropoff
customer_name	varchar(50)	Name of the person driving
customer_phone	varchar(20)	Contact number for immediate communication regarding the booking
emergency_contact_name	varchar(100)	Name of the emergency contact person
emergency_contact_phone	varchar(20)	Contact number of the emergency contact person
emergency_relationship	varchar(20)	Relationship to emergency contact to the customer
license_image	varchar(255)	Stores the file path or URL of the uploaded driving license image for verification purposes.

payment_receipt	varchar(255)	Stores the file path of the payment proof uploaded by the user.
custom_dropoff_address	text	Specific address details provided by the user when dropoff location is not Student Mall as default
total_rental_fee	decimal(8, 2)	Total calculated cost of the rental period before any discounts
deposit_amount	decimal(8, 2)	Security deposit amount collected to cover potential damages or fines
promo_code	varchar(255)	The specific promo code string applied to this booking,if any.
status	varchar(255)	Current state of booking. Values: 'Pending', 'Cancelled', 'Returning', 'Completed'
deposit_status	varchar(20)	Current status of the security deposit.
deposit_receipt_path	varchar(255)	Path to deposit refund receipt
deposit_returned_at	timestamp	Time when deposit was returned
rejection_reason	text	Reason if booking was rejected
payment_verified	tinyint(1)	Flag to indicate if the admin has verified the payment receipt (1 = Verified, 0 = Pending).
voucher_id	bigint	Foreign key. Reference vouchers table and links to the specific voucher used for discount tracking. Nullable
processed_by	bigint	Foreign Key. Identifies the Admin who approved or rejected the booking. Nullable because it remains empty (NULL) when the booking is first created and only gets populated once an Admin takes action.
refund_bank_name	varchar(60)	Name of the user's bank such as 'Maybank' to facilitate deposit refunds
refund_account_number	varchar(50)	User's bank account number for deposit refund transfer

refund_recipient_name	varchar(100)	Name of the account holder for the refund transaction
created_at	timestamp	Timestamp when the booking request was first created
updated_at	timestamp	Timestamp when the booking details or status were last modified

4.2.2.2 Entity: claims

Attribute Name	**Type**	**Description**
id	bigint	Primary key.Unique auto incrementation identifier for each claim records
matric_staff_id	varchar(15)	The staff ID of the individual making the claim.
user_id	bigint	Foreign key. References the users table to identify the claimant
claim_type	varchar(20)	The category/type of the claim (e.g., Fuel Reimbursement, Maintenance, Accident).
vehicle_plate	varchar(10)	The license plate number of the vehicle involved in the claim.
amount	decimal(10,2)	The total monetary value being claimed.
claim_date_time	datetime	The specific date and time when the incident or expense occurred.
description	text	Detailed explanation or notes regarding the reason for the claim.
status	enum('Pending', 'Approved', 'Rejected')	Current status of the claim processing (e.g., 'Pending', 'Approved', 'Rejected').
action_reason	text	The additional remark provided by the top manager for approving or rejecting the claim.
receipt_path	varchar(255)	Stores the file path or URL to the uploaded receipt image serving as proof of expense.

payment_proof	varchar(255)	Stores the file path or URL to the proof of payment (reimbursement) uploaded by the top manager.
processed_by	bigint	Foreign key. References the users table to identify who processes the claim. Nullable because it remains empty(NULL) when the claim is first created.
processed_at	timestamp	Timestamp indicating when the claim was processed.
created_at	timestamp	Timestamp when the claim record was created.
updated_at	timestamp	Timestamp when the claim record was last modified.

4.2.2.3 Entity: colleges

Attribute Name	Type	Description
id	bigint	Primary key.Unique auto incrementing identifier for each college
name	varchar(50)	The name of the college (e.g., KTDI, KTR).
created_at	timestamp	Timestamp indicating when the college record was created.
updated_at	timestamp	Timestamp indicating when the college record was last modified.

4.2.2.4 Entity: faculties

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for each faculty
name	varchar(100)	The official name of the faculty (e.g., Faculty of Computing).
created_at	timestamp	Timestamp indicating when the faculty record was created.
updated_at	timestamp	Timestamp indicating when the faculty record was last modified.

4.2.2.5 Entity: feedback

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for feedback record.
booking_id	bigint	Foreign key. References the bookings table to identify the rental transaction
ratings	json	JSON data storing detailed rating metrics (e.g., cleanliness score, vehicle condition score).
category	varchar(50)	The classification of the feedback (e.g., Mechanical Issue, Aircond Issue). Nullable.
description	text	Textual comment providing full details of the user's feedback or experience.
return_photo_url	varchar(255)	Stores the file path or URL of the photo taken upon vehicle return, used as evidence for vehicle condition.
created_at	timestamp	Timestamp indicating when the feedback was submitted.
updated_at	timestamp	Timestamp indicating when the feedback record was last modified.

4.2.2.6 Entity: fines

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for each fine.
booking_id	bigint	Foreign key. References the bookings table to identify the associated rental transaction.
amount	decimal(8, 2)	The monetary amount charged for the fine.
receipt_path	varchar(255)	Stores the file path or URL to the uploaded receipt showing proof that the fine has been paid.

reason	varchar(255)	The justification for issuing the fine (e.g., 'Late Return', 'Vehicle Damage', 'Smoking').
status	varchar(30)	The payment status of the fine (e.g., 'Unpaid', 'Paid').
paid_at	timestamp	Timestamp indicating when the fine payment was completed.
created_at	timestamp	Timestamp indicating when the fine record was created.
updated_at	timestamp	Timestamp indicating the last time the fine status or details were modified.

4.2.2.7 Entity: inspections

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for each inspection.
booking_id	bigint	Foreign key. References the bookings table to identify the associated rental transaction.
type	enum('pickup', 'return')	The stage of inspection, either 'pickup' (before rental) or 'return' (after rental).
fuel_level	int	The fuel gauge reading at the time of inspection (e.g., number of bars, 0-10).
mileage	int	The vehicle's odometer reading (total distance traveled) at the time of inspection.
remarks	text	General notes or comments made during the inspection.
photo_front	varchar(255)	Stores the file path or URL to the photo of the vehicle's front view.
photo_back	varchar(255)	Stores the file path or URL to the photo of the vehicle's rear view.
photo_left	varchar(255)	Stores the file path or URL to the photo of the vehicle's left side.

photo_right	varchar(255)	Stores the file path or URL to the photo of the vehicle's right side.
photo_dashboard	varchar(255)	Stores the file path or URL to the photo of the dashboard (fuel/mileage).
photo_keys	varchar(255)	Stores the file path or URL to the photo of the vehicle keys (usually for return inspection).
damage_photos	json	JSON array storing file paths/URLs to photos of specific damages found on the vehicle.
damage_description	text	Textual details describing any new or existing damage found on the vehicle.
created_by	bigint	Foreign key. References the users table to identify who performed the inspection.
created_at	timestamp	Timestamp indicating when the inspection was recorded.
updated_at	timestamp	Timestamp indicating when the inspection record was last modified.

4.2.2.8 Entity: loyalty_cards

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for each loyalty card
user_id	bigint	Foreign key. References the users table to identify the cardholder.
stamps	int	Current count of loyalty stamps accumulated by the user.
unread_stamps	int	Counter for new stamps that the user hasn't seen yet (used for notification badges/popups).
created_at	timestamp	Timestamp indicating when the loyalty card was first issued.

updated_at	timestamp	Timestamp indicating when the card details (e.g., stamp count) were last updated.

4.2.2.9 Entity: maintenance_logs

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for each maintenance log.
vehicle_id	bigint	Foreign key. References the vehicles table to identify the vehicle undergoing maintenance.
user_id	bigint	Foreign key. References the users table to identify who recorded the log.
date	date	The specific date when the maintenance was performed.
description	varchar(255)	Details of the maintenance work carried out (e.g., "Oil Change", "Tyre Replacement").
cost	decimal(8, 2)	The total cost incurred for the maintenance service.
created_at	timestamp	Timestamp indicating when the log entry was created in the system.
updated_at	timestamp	Timestamp indicating when the log entry was last modified.

4.2.2.10 Entity: notifications

Attribute Name	Type	Description
id	char(36)	Primary key. Unique identifier for notification
type	varchar(255)	Indicates the category of the notification such as distinguishing between a “Booking Approval” and a “Fine Issuance”
notifiable_type	varchar(255)	Identifies the type of user receiving the message

notifiable_id	bigint	The ID of the specific entity (user) receiving the notification.
data	text	JSON text containing the payload details of the notification (e.g., messages, links, related IDs).
read_at	timestamp	Timestamp indicating when the user read the notification; NULL if unread.
created_at	timestamp	Timestamp indicating when the notification was sent.
updated_at	timestamp	Timestamp indicating when the notification status was last updated such as marked as read.

4.2.2.11 Entity: password_reset_tokens

Attribute Name	Type	Description
email	varchar(60)	Primary key. The email address of the user requesting a password reset. Serves as the unique identifier for the request.
token	varchar(255)	A secure, hashed verification token generated by the system to validate the reset request.
created_at	timestamp	Timestamp indication when the reset request was created. Used to determine token expiration.

4.2.2.12 Entity: payments

Attribute Name	Type	Description
id	bigint	Primary key.Unique identifier for payment.
booking_id	bigint	Foreign key. References the bookings table to identify the associated rental transaction.
amount	decimal(8, 2)	The monetary value of the payment made.
payment_method	varchar(50)	The mode of payment (e.g., 'DuitNow QR', 'Manual Transfer').

receipt_image	varchar(255)	Stores the file path or URL to the uploaded proof of payment.
status	varchar(30)	Current status of the payment (e.g., 'Pending', 'Verified', 'Rejected').
created_at	timestamp	Timestamp indicating when the payment record was created.
updated_at	timestamp	Timestamp indicating when the payment status/details were last modified.

4.2.2.13 Entity: pricing_rules

Attribute Name	**Type**	**Description**
id	bigint	Primary key. Unique identifier for pricing rules
pricing_tier_id	bigint	Foreign key. References the pricing_tiers table to identify the associated pricing category.
hour_limit	int	The duration threshold (in hours) for this price point (e.g., 12 hours, 24 hours).
price	decimal(8, 2)	The rental cost associated with the specified hour limit.
created_at	timestamp	Timestamp indicating when this pricing rule was created.
updated_at	timestamp	Timestamp indicating when this pricing rule was last modified.

4.2.2.14 Entity: pricing_tiers

Attribute Name	**Type**	**Description**
id	bigint	Primary key. Unique auto incrementing identifier for pricing tiers.
name	varchar(50)	The name of the pricing category (e.g., "Tier 1", "Tier 2"). Assigned to vehicles to determine their rate structure.
description	text	Optional text describing the details or eligibility of this tier.

created_at	timestamp	Timestamp indicating when the pricing tier was created.
updated_at	timestamp	Timestamp indicating when the tier details were last modified.

4.2.2.15 Entity: promotions

Attribute Name	Type	Description
id	bigint	Primary key. Unique identifier for promotions
code	varchar(30)	The alphanumeric code users enter to redeem the promotion (e.g., "WELCOME10").
description	varchar(255)	Details explaining the promotion's benefits.
discount_value	decimal(8, 2)	The numerical value of the discount (can be percentage or fixed amount).
type	varchar(20)	The calculator type for the discount (e.g., 'percent', 'fixed', 'free_hours').
start_date	date	The date when the promotion becomes valid.
end_date	date	The date when the promotion expires.
is_active	tinyint(1)	Boolean flag (0 or 1) indicating if the promotion is currently enabled.
created_at	timestamp	Timestamp indicating when the promotion was created.
updated_at	timestamp	Timestamp indicating when the promotion was last modified.

4.2.2.16 Entity: users

Attribute Name	Type	Description
id	bigint	Primary key. Unique identifier for users
name	varchar(100)	The full name of the user.
email	varchar(60)	The user's email address (used for login and notifications).

email_verified_at	timestamp	Timestamp indicating when the user verified their email address.
password	varchar(255)	The hashed password for account security.
role	varchar(15)	The user's access level (e.g., 'customer', 'staff', 'top manager').
matric_staff_id	varchar(15)	Unique institutional ID (Matric No. for students, Staff ID for staff).
matric_card_path	varchar(255)	Stores the file path to the uploaded matric card image.
nric_passport_path	varchar(255)	Stores the file path to the uploaded NRIC or Passport image.
nric_passport	varchar(255)	The text value of the NRIC or Passport number.
expired_date	date	The expiration date of the user's license.
driving_license_path	varchar(255)	Stores the file path to the uploaded driving license image.
phone_number	varchar(20)	The user's contact phone number.
address	text	The residential or mailing address of the user.
nationality	varchar(100)	The nationality of the user.
emergency_name	varchar(100)	Name of the customer's emergency contact person.
emergency_contact	varchar(20)	Phone number of the customer's emergency contact person.
emergency_relationship	varchar(20)	The relationship of the emergency contact to the customer (e.g., Parent, Sibling).
is_blacklisted	tinyint(1)	Boolean flag indicating if the customer is banned from accessing the system.
blacklist_reason	text	Textual reason explaining why the user was blacklisted.
college_id	bigint	Foreign key. References the colleges table.
faculty_id	bigint	Foreign key. Reference the faculties table

remember_token	varchar(100)	Token used for "Remember Me" session functionality.
created_at	timestamp	Timestamp indicating when the user account was registered.
updated_at	timestamp	Timestamp indicating when the user profile was last updated.
bank_name	varchar(255)	Name of the bank for refund purposes.
account_name	varchar(255)	The bank account number for refunds or deposit
account_holder	varchar(255)	Name of the account holder as per bank records.
salary	decimal(10, 2)	The salary amount (relevant for staff/manager for internal records).

4.2.2.17 Entity: user_vouchers

Attribute Name	Type	Description
id	bigint	Primary key. Unique auto incrementing identifier for user_vouchers records.
user_id	varchar(15)	Foreign key. References the users table(matric_staff_id) to identify the voucher owner.
voucher_id	bigint	Foreign key. References the vouchers table to identify the specific voucher type
used_at	timestamp	Timestamp indicating when a voucher was applied to a booking. NULL if unused.
created_at	timestamp	Timestamp when the voucher was claimed by or assigned to the user.
updated_at	timestamp	Timestamp when the record was last modified such as marked as used.

4.2.2.18 Entity: vehicles

Attribute Name	Type	Description

id	bigint	Primary key. Unique auto incrementing identifier for each vehicle.
vehicle_id_custom	varchar(10)	Custom human readable identifier such as ‘V001’ used for internal fleet tracking
plate_number	varchar(10)	Official vehicle registration number. Unique system-wide.
brand	varchar(255)	Manufacturer of the vehicle such as ‘Perodua’
model	varchar(255)	Specific model name of the vehicle such as ‘Myvi’
year	int	Manufacturing year of the vehicle
capacity	int	Maximum number of passengers the vehicle can legally carry
price_per_hour	decimal(8, 2)	Base hourly rental rate. Used as a fallback if no specific tier is set because the price provided by stakeholder is 1hour, 3 hour and so on
vehicle_image	varchar(255)	File path to the uploaded image of the vehicle and save as URL
is_hasta_owned	tinyint(1)	Boolean flag 1 if the car is owned by Hasta, 0 if it is a third party rental
owner_name	varchar(100)	Name of the car owner
owner_ic_path	varchar(255)	File path to the uploaded IC copy of the car owner and save as URL
owner_license_path	varchar(255)	File path to the uploaded driving license copy of the car owner and save as URL
geran_path	varchar(255)	File path to the uploaded vehicle registration card and save as URL
insurance_cover_path	varchar(255)	File path to the insurance cover note document and save as URL
current_fuel_bars	int	Current fuel level recorded for tracking

status	varchar(30)	Current operational status. Values: ‘Available’, ‘Rented’, ‘Maintenance’, ‘Unavailable’
type_id	int	Identifier linking to vehicle type category
road_tax_expiry	date	Expiration date of the vehicle’s road tax
insurance_expiry	date	Expiration date of the vehicle insurance policy.
created_at	timestamp	Timestamp when the vehicle record was created
updated_at	timestamp	Timestamp when the vehicle details were last modified
pricing_tier_id	bigint	Foreign key links to pricing_tiers table

4.2.2.19 Entity: vehicle_types

Attribute Name	**Type**	**Description**
id	bigint	Primary key. Unique identifier for vehicle type
code	int	System code. Maps to the selection value in dropdown menus such as 1 for Compact, 2 for Sedan
name	varchar(30)	Type Name such as ‘Compact’, ‘Sedan’
base_price	decimal(8, 2)	Legacy field designated for base pricing. Retained for backward compatibility or future implementation, superseded by specific vehicle pricing rules.
created_at	timestamp	Timestamp when the record was created
updated_at	timestamp	Timestamp when the record was updated

4.2.2.20 Entity: vouchers

Attribute Name	Type	Description
id	bigint	Primary key. Unique identifier for voucher
user_id	varchar(15)	Foreign key links to users table(matric_staff_id). Nullable. If set, the voucher is exclusive to that specific user; if NULL, it is a public voucher
code	varchar(30)	Voucher code. The unique alphanumeric code users enter to redeem
name	varchar(50)	Display name of voucher. A simple descriptive label for the voucher
type	varchar(20)	Reward type. Logic determinant: ‘fixed’(RM) , ‘percent’(%), or free_hours
value	decimal(8, 2)	Reward value. Numerical amount corresponding to the type
is_active	tinyint(1)	Status flag. 1 indicates it is active or redeemable. Disabled or invalidated can be set by the admin.
expires_at	timestamp	Expiration timestamp. Nullable. If set, the voucher cannot be used after this date.
created_at	timestamp	Timestamp when the voucher was created
updated_at	timestamp	Timestamp when the voucher details were last modified.
single_use	tinyint(1)	One time use the flag. If 1, limit each unique user to redeeming this code only once
uses_remaining	int	Global usage counter. Nullable. Decrements on each use. If verified as greater than 0, limit total number of redemptions system-wide.

4.3 Normalization

It should be noted that the normalization approach adopted in this project focuses on the core transactional entities of the system, rather than normalizing the entire global ERD. The normalization process is centred on booking-related data, including user, vehicle, booking, and payment information, as these entities form the main operational workflow of the Hasta Travel Car Rental System. This approach allows the normalization steps from Unnormalized Form (UNF) to Third Normal Form (3NF) to be clearly illustrated, while ensuring that all attributes used are consistent with the ERD design. Other supporting entities are already well-structured in the ERD and are therefore not expanded further in this normalization discussion.

4.3.1 Unnormalized Form (UNF)

Initially, the system stores all booking-related information in one table, combining user, vehicle, booking, payment, and loyalty data. As a result, the table contains repeating groups and multi-valued attributes.

Functional Dependencies in UNF:

- booking_id → booking_date, pickup_date_time, return_date_time
- user_id → user_name, phone_number
- vehicle_id → model
- payment_id → payment_method, amount
- booking_id → receipt_image
- booking_id → ExtraCharge
- user_id → stamps

RENTAL_RECORD_UNF (booking_id, booking_date, user_id, user_name, phone_number, vehicle_id, model, pickup_date_time, return_date_time, payment_id, payment_method, amount, receipt_image {...}, ExtraCharge {...}, stamps {...})

Several issues can be observed in the unnormalized table. Some attributes contain multiple values such as ExtraCharge and receipt_image. In addition, user and vehicle information is repeatedly stored, as different entities are combined into a single table. This structure may lead to data anomalies during update, insertion, and deletion operations.

4.3.2 First Normal Form (1NF)

To achieve First Normal Form (1NF), all repeating groups are removed and each attribute is ensured to contain atomic values. The multi-valued attributes (ExtraCharge, receipt_image) are moved to separate tables or flattened, leaving the main relation with atomic columns.

Functional Dependencies in BOOKING_1NF:

- booking_id → booking_date, pickup_date_time, return_date_time, user_id, vehicle_id, payment_id
- user_id → user_name, phone_number
- vehicle_id → model
- payment_id → payment_method, amount

Relation:

BOOKING_1NF (booking_id PK, booking_date, user_id, user_name, phone_number, vehicle_id, model, pickup_date_time, return_date_time, payment_id, payment_method, amount**)**

Explanation:

After applying 1NF, all attributes contain atomic values and repeating groups have been removed. However, user, vehicle, and payment details are still stored together in the same table, which may lead to redundancy and dependency issues that will be addressed in the next normalization stages.

4.3.3 Second Normal Form (2NF)

Second Normal Form (2NF) requires that the relation is in 1NF and contains no partial dependencies. A partial dependency occurs when a non-key attribute depends on only a part of a composite primary key.

The Primary Key of BOOKING_1NF is a single attribute: booking_id. Since the primary key is not composite, it is impossible for any attribute to be dependent on "part" of the key. All non-key attributes are fully dependent on the entire primary key.

Relation:

BOOKING_2NF (booking_id PK, booking_date, user_id, user_name, phone_number, vehicle_id, model, pickup_date_time, return_date_time, payment_id, payment_method, amount)

Explanation:

The relation is automatically in 2NF because it possesses a single-attribute primary key. The redundancies related to User and Vehicle data still exist, but they are identified as Transitive Dependencies, which will be addressed in the next stage (3NF).

4.3.4 Third Normal Form (3NF)

Third Normal Form (3NF) is achieved by eliminating transitive dependencies, where a non-key attribute depends on another non-key attribute rather than the primary key.

Identified Transitive Dependencies:

- booking_id → user_id → user_name, phone_number
- user_id → college_id → college_name
- booking_id → vehicle_id → model
- vehicle_id → pricing_tier_id → hourly_rate
- booking_id → payment_id → amount, payment_method

Decomposition into 3NF: We decompose the 2NF relation by creating separate tables for these dependencies to ensure that every non-key attribute depends only on the Primary Key.

Relations:

users (<u>user_id PK</u>, user_name, phone_number, college_id FK)

colleges (<u>college_id PK</u>, college_name)

vehicles (<u>vehicle_id PK</u>, model, pricing_tier_id FK)

pricing_tiers (<u>pricing_tier_id PK</u>, tier_name, hourly_rate)

payments (<u>payment_id PK</u>, booking_id FK, payment_method, amount)

bookings (<u>booking_id PK</u>, user_id FK, vehicle_id FK, pickup_date_time, return_date_time, booking_date, booking_status)

Explanation:

All transitive dependencies have been removed. The User and College details are separated into their own tables. The Vehicle and Pricing details are separated. The Payment details are separated. The main Bookings table now contains only foreign keys referencing these entities. Each non-key attribute now depends strictly on the primary key of its own table.

4.3.5 Normalization Summary

In summary, the final database design meets the requirements of Third Normal Form (3NF). The normalization process has minimized data redundancy and reduced the possibility of update, insertion, and deletion anomalies.

BCNF Consideration:

The relations were further evaluated against Boyce-Codd Normal Form (BCNF). In the proposed design, every determinant (the left side of a functional dependency) is a candidate key in its respective table. As no BCNF violations were identified, further decomposition was not required. Therefore, the database design remains in Third Normal Form (3NF), which is sufficient for the system requirements.

5.0 Relational DB Schemas (after normalization)

The following relational schemas represent the final structure of the core transactional entities after applying normalization rules up to Third Normal Form (3NF).These schemas reflect only the booking-related entities (User, Vehicle, Booking, Payment, College, and Pricing Tier) analyzed in Section 4.3, rather than the entire global database. Supporting entities (e.g., Inspections, Maintenance) are excluded here as they were not part of this specific normalization process.

colleges (college_id PK, college_name)

users (user_id PK, user_name, phone_number, college_id FK)

pricing_tiers (pricing_tier_id PK, tier_name, hourly_rate)

vehicles (vehicle_id PK, model, pricing_tier_id FK)

bookings (booking_id PK, user_id FK, vehicle_id FK, pickup_date_time, return_date_time, booking_date, booking_status)

payments (payment_id PK, booking_id FK, payment_method, amount)

6.0 SQL Statements (DDL & DML)

6.1 Bookings Table

Create Table

```
CREATE TABLE bookings (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id bigint UNSIGNED NOT NULL,
  vehicle_id bigint UNSIGNED NOT NULL,
  pickup_date_time datetime NOT NULL,
  return_date_time datetime NOT NULL,
  pickup_location varchar(50) DEFAULT 'Student Mall',
  custom_pickup_address text,
  dropoff_location varchar(50) DEFAULT 'Student Mall',
  customer_name varchar(50) NOT NULL,
  customer_phone varchar(20) NOT NULL,
  emergency_contact_name varchar(100) NOT NULL,
  emergency_contact_phone varchar(20) NOT NULL,
  emergency_relationship varchar(20) NOT NULL,
  license_image varchar(255) NOT NULL,
  payment_receipt varchar(255) NOT NULL,
  custom_dropoff_address text,
  total_rental_fee decimal(8,2) NOT NULL,
  deposit_amount decimal(8,2) DEFAULT 0.00,
  promo_code varchar(255) NULL,
  status varchar(255) DEFAULT 'Pending',
  deposit_status varchar(20) NOT NULL,
  deposit_receipt_path varchar(255) NOT NULL,
  deposit_returned_at timestamp,
  rejection_reason text,
  payment_verified tinyint(1) DEFAULT 0,
  voucher_id bigint,
  processed_by bigint,
  refund_bank_name varchar(60) NOT NULL,
  refund_account_number varchar(50) NOT NULL,
  refund_recipient_name varchar(100) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE,
  FOREIGN KEY (vehicle_id) REFERENCES vehicles (id) ON DELETE CASCADE
);
```

Insert 20 Values into Table

```
INSERT INTO bookings (user_id, vehicle_id, pickup_date_time, return_date_time,
pickup_location, dropoff_location, customer_name, customer_phone,
emergency_contact_name, emergency_contact_phone, emergency_relationship,
license_image, payment_receipt, total_rental_fee, deposit_amount, status,
```

```
deposit_status, deposit_receipt_path, refund_bank_name, refund_account_number,
refund_recipient_name)
VALUES
    (1, 1, '2025-10-01 10:00:00', '2025-10-01 14:00:00', 'Student Mall', 'Student
Mall', 'Ali Bin Abu', '0123456789', 'Abu Bakar', '0198765432', 'Father',
'licenses/ali.jpg', 'receipts/pay_001.jpg', 32.00, 50.00, 'Completed', 'Returned',
'receipts/dep_001.jpg', 'Maybank', '1122334455', 'Ali Bin Abu'),
    (2, 2, '2025-10-02 09:00:00', '2025-10-02 18:00:00', 'Kolej Tub Mohammed',
'Student Mall', 'Siti Sarah', '0134567890', 'Sarah Mother', '0187654321',
'Mother', 'licenses/siti.jpg', 'receipts/pay_002.jpg', 108.00, 100.00,
'Completed', 'Returned', 'receipts/dep_002.jpg', 'CIMB', '2233445566', 'Siti
Sarah'),
    (3, 1, '2025-10-05 12:00:00', '2025-10-05 16:00:00', 'Student Mall',
'Library', 'Chong Wei', '0145678901', 'Chong Father', '0176543210', 'Father',
'licenses/chong.jpg', 'receipts/pay_003.jpg', 32.00, 50.00, 'Pending', 'Pending',
'receipts/dep_003.jpg', 'Public Bank', '3344556677', 'Chong Wei'),
    (1, 3, '2025-11-20 08:00:00', '2025-11-20 20:00:00', 'Student Mall', 'Student
Mall', 'Ali Bin Abu', '0123456789', 'Abu Bakar', '0198765432', 'Father',
'licenses/ali.jpg', 'receipts/pay_004.jpg', 120.00, 100.00, 'Confirmed', 'Paid',
'receipts/dep_004.jpg', 'Maybank', '1122334455', 'Ali Bin Abu'),
    (2, 4, '2025-10-08 14:00:00', '2025-10-08 18:00:00', 'Kolej Perdana', 'Kolej
Perdana', 'Siti Sarah', '0134567890', 'Sarah Mother', '0187654321', 'Mother',
'licenses/siti.jpg', 'receipts/pay_005.jpg', 60.00, 50.00, 'Cancelled',
'Refunded', 'receipts/dep_005.jpg', 'CIMB', '2233445566', 'Siti Sarah'),
    (5, 5, '2025-10-25 10:00:00', '2025-10-26 10:00:00', 'Student Mall', 'Student
Mall', 'Muthu Sami', '0167890123', 'Sami Father', '0154321098', 'Father',
'licenses/muthu.jpg', 'receipts/pay_006.jpg', 432.00, 200.00, 'Active', 'Paid',
'receipts/dep_006.jpg', 'RHB', '4455667788', 'Muthu Sami'),
    (3, 2, '2025-10-09 09:00:00', '2025-10-09 12:00:00', 'Student Mall', 'Kolej
9', 'Chong Wei', '0145678901', 'Chong Father', '0176543210', 'Father',
'licenses/chong.jpg', 'receipts/pay_007.jpg', 36.00, 50.00, 'Completed',
'Returned', 'receipts/dep_007.jpg', 'Public Bank', '3344556677', 'Chong Wei'),
    (4, 1, '2025-10-10 10:00:00', '2025-10-10 11:00:00', 'Student Mall', 'Student
Mall', 'New User', '0111234567', 'User Mom', '0117654321', 'Mother',
'licenses/invalid.jpg', 'receipts/pay_008.jpg', 8.00, 50.00, 'Rejected',
'Refunded', 'receipts/dep_008.jpg', 'Bank Islam', '5566778899', 'New User'),
    (5, 3, '2025-11-25 08:00:00', '2025-11-26 08:00:00', 'Student Mall', 'Student
Mall', 'Muthu Sami', '0167890123', 'Sami Father', '0154321098', 'Father',
'licenses/muthu.jpg', 'receipts/pay_009.jpg', 240.00, 150.00, 'Confirmed', 'Paid',
'receipts/dep_009.jpg', 'RHB', '4455667788', 'Muthu Sami'),
    (2, 5, '2025-10-15 10:00:00', '2025-10-15 12:00:00', 'Student Mall', 'F
Computing', 'Siti Sarah', '0134567890', 'Sarah Mother', '0187654321', 'Mother',
'licenses/siti.jpg', 'receipts/pay_010.jpg', 36.00, 50.00, 'Completed',
'Returned', 'receipts/dep_010.jpg', 'CIMB', '2233445566', 'Siti Sarah'),
    (1, 4, '2025-12-01 08:00:00', '2025-12-01 12:00:00', 'Kolej 10', 'Student
Mall', 'Ali Bin Abu', '0123456789', 'Abu Bakar', '0198765432', 'Father',
'licenses/ali.jpg', 'receipts/pay_011.jpg', 60.00, 50.00, 'Pending', 'Pending',
'receipts/dep_011.jpg', 'Maybank', '1122334455', 'Ali Bin Abu'),
    (3, 1, '2025-10-18 14:00:00', '2025-10-18 16:00:00', 'Student Mall', 'Student
Mall', 'Chong Wei', '0145678901', 'Chong Father', '0176543210', 'Father',
```

```
'licenses/chong.jpg', 'receipts/pay_012.jpg', 16.00, 50.00, 'Completed',
'Returned', 'receipts/dep_012.jpg', 'Public Bank', '3344556677', 'Chong Wei'),
    (5, 2, '2025-10-20 09:00:00', '2025-10-20 18:00:00', 'Student Mall', 'Student
Mall', 'Muthu Sami', '0167890123', 'Sami Father', '0154321098', 'Father',
'licenses/muthu.jpg', 'receipts/pay_13.jpg', 108.00, 100.00, 'Cancelled',
'Refunded', 'receipts/dep_13.jpg', 'RHB', '4455667788', 'Muthu Sami'),
    (2, 3, '2025-12-05 10:00:00', '2025-12-05 14:00:00', 'Student Mall', 'Student
Mall', 'Siti Sarah', '0134567890', 'Sarah Mother', '0187654321', 'Mother',
'licenses/siti.jpg', 'receipts/pay_14.jpg', 40.00, 50.00, 'Confirmed', 'Paid',
'receipts/dep_14.jpg', 'CIMB', '2233445566', 'Siti Sarah'),
    (4, 5, '2025-10-22 08:00:00', '2025-10-22 20:00:00', 'Skudai', 'Student Mall',
'New User', '0111234567', 'User Mom', '0117654321', 'Mother',
'licenses/newuser.jpg', 'receipts/pay_15.jpg', 216.00, 200.00, 'Completed',
'Returned', 'receipts/dep_15.jpg', 'Bank Islam', '5566778899', 'New User'),
    (1, 2, '2025-12-10 12:00:00', '2025-12-10 16:00:00', 'Student Mall', 'Student
Mall', 'Ali Bin Abu', '0123456789', 'Abu Bakar', '0198765432', 'Father',
'licenses/ali.jpg', 'receipts/pay_16.jpg', 48.00, 50.00, 'Pending', 'Pending',
'receipts/dep_16.jpg', 'Maybank', '1122334455', 'Ali Bin Abu'),
    (5, 1, '2025-10-28 09:00:00', '2025-10-28 11:00:00', 'Student Mall', 'Student
Mall', 'Muthu Sami', '0167890123', 'Sami Father', '0154321098', 'Father',
'licenses/muthu.jpg', 'receipts/pay_17.jpg', 16.00, 50.00, 'Completed',
'Returned', 'receipts/dep_17.jpg', 'RHB', '4455667788', 'Muthu Sami'),
    (3, 4, '2025-10-30 10:00:00', '2025-10-30 14:00:00', 'Student Mall', 'Student
Mall', 'Chong Wei', '0145678901', 'Chong Father', '0176543210', 'Father',
'licenses/chong.jpg', 'receipts/pay_fake.jpg', 60.00, 50.00, 'Rejected',
'Pending', 'receipts/dep_fake.jpg', 'Public Bank', '3344556677', 'Chong Wei'),
    (4, 2, '2025-12-15 08:00:00', '2025-12-15 20:00:00', 'Student Mall', 'Student
Mall', 'New User', '0111234567', 'User Mom', '0117654321', 'Mother',
'licenses/newuser.jpg', 'receipts/pay_19.jpg', 144.00, 100.00, 'Confirmed',
'Paid', 'receipts/dep_19.jpg', 'Bank Islam', '5566778899', 'New User'),
    (1, 5, '2025-11-01 10:00:00', '2025-11-01 15:00:00', 'Student Mall', 'Kolej
11', 'Ali Bin Abu', '0123456789', 'Abu Bakar', '0198765432', 'Father',
'licenses/ali.jpg', 'receipts/pay_20.jpg', 90.00, 50.00, 'Completed', 'Returned',
'receipts/dep_20.jpg', 'Maybank', '1122334455', 'Ali Bin Abu');
```

6.2 Claims Table

Create Table

```
CREATE TABLE claims (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  matric_staff_id varchar(15) NOT NULL,
  user_id bigint UNSIGNED NOT NULL,
  claim_type varchar(20) NOT NULL,
  vehicle_plate varchar(10)
  amount decimal(10,2) NOT NULL,
  claim_date_time datetime NOT NULL,
  description text,
  status enum('Pending', 'Approved', 'Rejected') DEFAULT 'Pending',
  action_reason text NOT NULL,
  receipt_path varchar(255) NULL,
  payment_proof varchar(255) NULL,
  processed_by bigint NOT NULL,
  processed_at timestamp NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (user_id) REFERENCES users (id)
);
```

Insert 20 Values into Table

```
INSERT INTO claims (matric_staff_id, user_id, claim_type, vehicle_plate, amount,
claim_date_time, description, status, action_reason, receipt_path, payment_proof`,
processed_by, processed_at)
VALUES
    ('S001', 4, 'Maintenance', 'JKA1234', 120.00, '2025-10-01 09:00:00', 'Routine
oil change and filter replacement', 'Approved', 'Scheduled maintenance due',
'staff_claims/oil_01.jpg', 'proof/p01.jpg', 4, NOW()),
    ('S002', 4, 'Fuel', 'WB4567', 50.00, '2025-10-02 08:30:00', 'Top up fuel for
rental readiness', 'Approved', 'Vehicle low on fuel', 'staff_claims/fuel_02.jpg',
'proof/p02.jpg', 4, NOW()),
    ('S001', 4, 'Maintenance', 'VGH8901', 400.00, '2025-10-05 14:00:00', 'Replaced
2 worn front tyres', 'Approved', 'Safety hazard identified',
'staff_claims/tyre_03.jpg', 'proof/p03.jpg', 4, NOW()),
    ('S003', 4, 'Cleaning', 'JQV3322', 15.00, '2025-10-06 10:00:00', 'External
wash and vacuum', 'Pending', '', 'staff_claims/wash_04.jpg', NULL, 4, NOW()),

    ('S001', 4, 'Repair', 'BRA5555', 250.00, '2025-10-10 11:30:00', 'Replaced dead
battery on site', 'Approved', 'Vehicle wouldn''t start',
'staff_claims/batt_05.jpg', 'proof/p05.jpg', 4, NOW()),
    ('S002', 4, 'Repair', 'JKA1234', 150.00, '2025-10-12 16:00:00', 'Polishing and
touch-up for bumper scratch', 'Approved', 'Restoring vehicle condition',
'staff_claims/repair_06.jpg', 'proof/p06.jpg', 4, NOW()),
```

```
    ('S001', 4, 'Legal', 'WB4567', 50.00, '2025-10-15 09:00:00', 'Puspakom
inspection fee', 'Approved', 'Annual inspection requirement',
'staff_claims/leg_07.jpg', 'proof/p07.jpg', 4, NOW()),
    ('S003', 4, 'Fuel', 'JQV3322', 80.00, '2025-10-16 19:00:00', 'Full tank
refuel', 'Rejected', 'Reason unclear/Not authorized', 'staff_claims/fuel_08.jpg',
NULL, 4, NOW()),
    ('S001', 4, 'Maintenance', 'VGH8901', 180.00, '2025-10-20 10:00:00', 'Aircon
gas refill and cleaning', 'Pending', '', 'staff_claims/ac_09.jpg', NULL, 4,
NOW()),
    ('S002', 4, 'Cleaning', 'BRA5555', 20.00, '2025-10-22 14:00:00', 'Deep clean
interior', 'Approved', 'Customer complaint regarding smell',
'staff_claims/clean_10.jpg', 'proof/p10.jpg', 4, NOW()),
    ('S001', 4, 'Repair', 'JKA1234', 80.00, '2025-10-25 09:00:00', 'Fix loose side
mirror', 'Approved', 'Safety check', 'staff_claims/c11.jpg', 'proof/p11.jpg', 4,
NOW()),
    ('S002', 4, 'Fuel', 'VGH8901', 40.00, '2025-10-26 08:00:00', 'Refuel before
booking #105', 'Approved', 'Prep for customer', 'staff_claims/c12.jpg',
'proof/p12.jpg', 4, NOW()),
    ('S003', 4, 'Maintenance', 'WB4567', 60.00, '2025-10-28 11:00:00', 'Alignment
and balancing', 'Pending', '', 'staff_claims/c13.jpg', NULL, 4, NOW()),
    ('S001', 4, 'Parts', 'JQV3322', 30.00, '2025-10-30 13:00:00', 'Buy new floor
mats', 'Approved', 'Old ones were torn', 'staff_claims/c14.jpg', 'proof/p14.jpg',
4, NOW()),
    ('S002', 4, 'Repair', 'BRA5555', 120.00, '2025-11-01 15:00:00', 'Replace
broken tail light bulb', 'Approved', 'Burnt out', 'staff_claims/c15.jpg',
'proof/p15.jpg', 4, NOW()),
    ('S001', 4, 'Towing', 'JKA1234', 150.00, '2025-11-03 18:00:00', 'Tow truck
from highway breakdown', 'Pending', '', 'staff_claims/c16.jpg', NULL, 4, NOW()),
    ('S003', 4, 'Cleaning', 'VGH8901', 15.00, '2025-11-05 09:00:00', 'Standard
wash', 'Approved', 'Weekly wash', 'staff_claims/c17.jpg', 'proof/p17.jpg', 4,
NOW()),
    ('S002', 4, 'Fuel', 'WB4567', 45.00, '2025-11-07 08:30:00', 'Refuel',
'Approved', 'Vehicle return empty', 'staff_claims/c18.jpg', 'proof/p18.jpg', 4,
NOW()),
    ('S001', 4, 'Maintenance', 'JQV3322', 90.00, '2025-11-10 10:00:00', 'Brake pad
inspection & cleaning', 'Approved', 'Squeaky brakes', 'staff_claims/c19.jpg',
'proof/p19.jpg', 4, NOW()),
    ('S002', 4, 'Other', 'BRA5555', 10.00, '2025-11-12 14:00:00', 'Parking fee',
'Rejected', 'Staff parking is free', 'staff_claims/c20.jpg', NULL, 4, NOW());
```

6.3 Colleges Table

Create Table

```
CREATE TABLE colleges (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  name varchar(50) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert Default Values into Table

```
INSERT INTO colleges (name)
VALUES
    ('KLG'),
    ('Outside Campus'),
    ('KTC'),
    ('KTHO'),
    ('KDOJ'),
    ('KDSE'),
    ('KP'),
    ('KTR'),
    ('KTDI'),
    ('KRP'),
    ('KTF'),
    ('K9K10');
```

6.4 Faculties Table

Create Table

```
CREATE TABLE faculties (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  name varchar(100) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert Default Values into Table

```
INSERT INTO faculties (name)
VALUES
    ('Not Applicable'),
    ('Faculty of Built Environment and Surveying'),
    ('Faculty of Artificial Intelligence'),
    ('Faculty of Civil Engineering'),
    ('Faculty of Electrical Engineering'),
    ('Faculty of Chemical Engineering'),
    ('Faculty of Chemical and Energy Engineering'), -- Often distinct entities in
history
    ('Faculty of Mechanical Engineering'),
    ('Faculty of Computing'),
    ('Faculty of Management'),
    ('Faculty of Science'),
    ('Faculty of Educational Sciences and Technology'),
    ('Faculty of Social Sciences and Humanities');
```

6.5 Feedback Table

Create Table

```
CREATE TABLE feedback (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  booking_id bigint UNSIGNED NULL,
  ratings int,
  category varchar(50),
  description text NOT NULL,
  return_photo_url varchar(255) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (booking_id) REFERENCES bookings (id)
);
```

Insert 20 Values into Table

```
INSERT INTO feedback (booking_id, ratings, category, description,
return_photo_url)
VALUES
    (1, 5, 'Service', 'Car was clean and staff was polite.',
'feedback/photo_01.jpg'),
    (2, 4, 'Vehicle Condition', 'Car runs smooth, but aircon took time to cool.',
'feedback/photo_02.jpg'),
    (3, 3, 'Cleanliness', 'Interior was bit dusty, but overall okay.',
'feedback/photo_03.jpg'),
    (1, 5, 'Overall Experience', 'Smooth process from pickup to return.',
'feedback/photo_04.jpg'),
    (4, 4, 'Suggestion', 'Maybe add phone holder in the cars.',
'feedback/photo_05.jpg'),
    (2, 2, 'Vehicle Condition', 'Wipers were making noise during rain.',
'feedback/photo_06.jpg'),
    (5, 5, 'Staff', 'Mr. Ali was very helpful during inspection.',
'feedback/photo_07.jpg'),
    (3, 4, 'Pricing', 'Affordable rates for students.', 'feedback/photo_08.jpg'),
    (1, 3, 'Convenience', 'Pickup point was a bit far from my college.',
'feedback/photo_09.jpg'),
    (6, 5, 'Service', 'Staff was understanding about my 10 min delay.',
'feedback/photo_10.jpg'),
    (2, 5, 'Vehicle Condition', 'Car given with full tank, appreciated.',
'feedback/photo_11.jpg'),
    (4, 1, 'Cleanliness', 'Found trash under the seat.', 'feedback/photo_12.jpg'),
    (5, 4, 'App Usability', 'Booking through website was easy.',
'feedback/photo_13.jpg'),
    (1, 5, 'Driving Experience', 'Perodua Axia is perfect for campus drive.',
'feedback/photo_14.jpg'),
    (3, 2, 'Vehicle Condition', 'Tyre pressure warning light was on.',
'feedback/photo_15.jpg'),
```

```
    (6, 5, 'Service', 'Inspection took only 5 mins. Fast!',
'feedback/photo_16.jpg'),
    (2, 3, 'Cleanliness', 'Car had a faint cigarette smell.',
'feedback/photo_17.jpg'),
    (4, 5, 'Overall Experience', 'Will definitely rent again next week.',
'feedback/photo_18.jpg'),
    (5, 4, 'App Usability', 'Payment gateway was a bit slow, but worked.',
'feedback/photo_19.jpg'),
    (1, 5, 'Overall Experience', 'Best rental service in UTM.',
'feedback/photo_20.jpg');
```

6.6 Fines Table

Create Table

```
CREATE TABLE fines (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  booking_id bigint UNSIGNED NOT NULL,
  amount decimal(8,2) NOT NULL,
  receipt_path varchar(255) NOT NULL,
  reason varchar(255) NOT NULL,
  status varchar(30) DEFAULT 'Unpaid',
  paid_at timestamp,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (booking_id) REFERENCES bookings (id) ON DELETE CASCADE
);
```

Insert 20 Values into Table

```
INSERT INTO fines (booking_id, amount, receipt_path, reason, status, paid_at)
VALUES
    (1, 15.00, '', 'Returned 1 hour late', 'Unpaid', NULL),
    (2, 50.00, 'fines/rec_01.jpg', 'Smoking detected in vehicle', 'Paid',
'2025-10-02 18:30:00'),
    (3, 30.00, '', 'Excessive sand/mud in interior', 'Unpaid', NULL),
    (4, 150.00, 'fines/rec_02.jpg', 'Minor scratch on rear bumper', 'Paid',
'2025-11-20 20:15:00'),
    (5, 20.00, '', 'Lost parking pass provided with car', 'Unpaid', NULL),
    (6, 25.00, 'fines/rec_03.jpg', 'Returned with less than half tank', 'Paid',
'2025-10-26 10:15:00'),
    (1, 150.00, '', 'Returned 1 day late', 'Unpaid', NULL),
    (2, 80.00, 'fines/rec_04.jpg', 'Drink spill on passenger seat', 'Paid',
'2025-10-15 12:10:00'),
    (3, 300.00, '', 'Speeding ticket received during rental', 'Unpaid', NULL),
    (4, 250.00, 'fines/rec_05.jpg', 'Replacement cost for lost key fob', 'Paid',
'2025-10-22 20:30:00'),
    (5, 10.00, '', 'Returned 30 mins late', 'Unpaid', NULL),
    (6, 40.00, 'fines/rec_06.jpg', 'Interior smells of durian', 'Paid',
'2025-10-28 11:15:00'),
    (1, 100.00, '', 'Dent on driver door', 'Unpaid', NULL),
    (2, 50.00, 'fines/rec_07.jpg', 'Cracked side mirror casing', 'Paid',
'2025-10-09 12:30:00'),
    (3, 20.00, '', 'Missing car mats', 'Unpaid', NULL),
    (4, 35.00, 'fines/rec_08.jpg', 'Low fuel level', 'Paid', '2025-11-01
15:15:00'),
    (5, 500.00, '', 'Accident damage excess fee', 'Unpaid', NULL),
    (6, 60.00, 'fines/rec_09.jpg', 'Special cleaning fee', 'Paid', '2025-12-05
14:15:00'),
    (1, 15.00, '', 'Returned late (Traffic)', 'Unpaid', NULL),
```

```
    (2, 200.00, 'fines/rec_10.jpg', 'Illegal parking clamp fee', 'Paid',
'2025-12-15 20:15:00');
```

6.7 Inspections Table

Create Table

```
CREATE TABLE inspections (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  booking_id bigint UNSIGNED NOT NULL,
  type enum('pickup','return') NOT NULL,
  fuel_level int NOT NULL,
  mileage int NOT NULL,
  remarks text NULL,
  photo_front varchar(255) NOT NULL,
  photo_back varchar(255) NOT NULL,
  photo_left varchar(255) NOT NULL,
  photo_right varchar(255) NOT NULL,
  photo_dashboard varchar(255) NOT NULL,
  photo_keys varchar(255) NOT NULL,
  damage_photos varchar(255),
  damage_description text,
  created_by bigint UNSIGNED NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (booking_id) REFERENCES bookings (id) ON DELETE CASCADE,
  FOREIGN KEY (created_by) REFERENCES users (id)
);
```

Insert 20 Values into Table

```
INSERT INTO inspections (booking_id, type, fuel_level, mileage, remarks,
photo_front, photo_back, photo_left, photo_right, photo_dashboard, photo_keys,
damage_photos, damage_description, created_by)
VALUES
    (1, 'pickup', 5, 20100, 'All good condition', 'insp/1_f.jpg', 'insp/1_b.jpg',
'insp/1_l.jpg',   'insp/1_r.jpg', 'insp/1_d.jpg', 'insp/1_k.jpg',  NULL, NULL, 4),
    (1, 'return', 4, 20150, 'Returned clean', 'insp/2_f.jpg', 'insp/2_b.jpg',
'insp/2_l.jpg', 'insp/2_r.jpg', 'insp/2_d.jpg', 'insp/2_k.jpg', NULL, NULL, 4),
    (2, 'pickup', 5, 15000, 'Existing scratch on left bumper', 'insp/3_f.jpg',
'insp/3_b.jpg', 'insp/3_l.jpg', 'insp/3_r.jpg', 'insp/3_d.jpg', 'insp/3_k.jpg',
'insp/3_dmg.jpg', 'Minor scratch on left front bumper', 4),
    (2, 'return', 5, 15200, 'No new damage', 'insp/4_f.jpg', 'insp/4_b.jpg',
'insp/4_l.jpg',  'insp/4_r.jpg', 'insp/4_d.jpg', 'insp/4_k.jpg', 'insp/3_dmg.jpg',
'Same scratch as pickup', 4),
    (3, 'pickup', 3, 30500, 'Handed over with 3 bars fuel', 'insp/5_f.jpg',
'insp/5_b.jpg', 'insp/5_l.jpg', 'insp/5_r.jpg', 'insp/5_d.jpg', 'insp/5_k.jpg',
NULL, NULL, 4),
    (3, 'return', 3, 30650, 'Dent found on rear right door', 'insp/6_f.jpg',
'insp/6_b.jpg', 'insp/6_l.jpg', 'insp/6_r.jpg', 'insp/6_d.jpg', 'insp/6_k.jpg',
'insp/6_dmg.jpg', 'New dent on rear right door', 4),
    (4, 'pickup', 5, 5000, 'New car condition', 'insp/7_f.jpg', 'insp/7_b.jpg',
'insp/7_l.jpg', 'insp/7_r.jpg', 'insp/7_d.jpg', 'insp/7_k.jpg', NULL, NULL, 4),
```

```
    (4, 'return', 5, 5300, 'Interior extremely dirty/muddy',  'insp/8_f.jpg',
'insp/8_b.jpg', 'insp/8_l.jpg', 'insp/8_r.jpg', 'insp/8_d.jpg', 'insp/8_k.jpg',
NULL, 'Mud all over floor mats and seats', 4),
    (5, 'pickup', 5, 45000, 'Old vehicle, many minor scratches',  'insp/9_f.jpg',
'insp/9_b.jpg', 'insp/9_l.jpg', 'insp/9_r.jpg', 'insp/9_d.jpg', 'insp/9_k.jpg',
'insp/9_dmg.jpg', 'Multiple existing scratches documented', 4),
    (5, 'return', 4, 45150, 'No fresh damage', 'insp/10_f.jpg', 'insp/10_b.jpg',
'insp/10_l.jpg', 'insp/10_r.jpg', 'insp/10_d.jpg', 'insp/10_k.jpg', NULL, NULL,
4),
    (6, 'pickup', 5, 12000, 'Good', 'i/11f.jpg', 'i/11b.jpg', 'i/11l.jpg',
'i/11r.jpg', 'i/11d.jpg', 'i/11k.jpg', NULL, NULL, 4),
    (6, 'return', 5, 12050, 'Good', 'i/12f.jpg', 'i/12b.jpg', 'i/12l.jpg',
'i/12r.jpg', 'i/12d.jpg', 'i/12k.jpg', NULL, NULL, 4),
    (1, 'pickup', 5, 22000, 'Re-rented immediately', 'i/13f.jpg', 'i/13b.jpg',
'i/13l.jpg', 'i/13r.jpg', 'i/13d.jpg', 'i/13k.jpg', NULL, NULL, 4),
    (2, 'pickup', 5, 16000, 'OK', 'i/14f.jpg', 'i/14b.jpg', 'i/14l.jpg',
'i/14r.jpg', 'i/14d.jpg', 'i/14k.jpg', NULL, NULL, 4),
    (2, 'return', 2, 16400, 'Returned with low fuel', 'i/15f.jpg', 'i/15b.jpg',
'i/15l.jpg', 'i/15r.jpg', 'i/15d.jpg', 'i/15k.jpg', NULL, 'Fuel bar blinking', 4),
    (3, 'pickup', 5, 31000, 'OK', 'i/16f.jpg', 'i/16b.jpg', 'i/16l.jpg',
'i/16r.jpg', 'i/16d.jpg', 'i/16k.jpg', NULL, NULL, 4),
    (4, 'pickup', 5, 6000, 'OK', 'i/17f.jpg', 'i/17b.jpg', 'i/17l.jpg',
'i/17r.jpg', 'i/17d.jpg', 'i/17k.jpg', NULL, NULL, 4),
    (5, 'return', 5, 46000, 'Lost hubcap', 'i/18f.jpg', 'i/18b.jpg', 'i/18l.jpg',
'i/18r.jpg', 'i/18d.jpg', 'i/18k.jpg', 'i/18dmg.jpg', 'Front left hubcap missing',
4),
    (6, 'pickup', 5, 13000, 'OK', 'i/19f.jpg', 'i/19b.jpg', 'i/19l.jpg',
'i/19r.jpg', 'i/19d.jpg', 'i/19k.jpg', NULL, NULL, 4),
    (1, 'return', 5, 22100, 'Smokes smell', 'i/20f.jpg', 'i/20b.jpg', 'i/20l.jpg',
'i/20r.jpg', 'i/20d.jpg', 'i/20k.jpg', NULL, 'Strong cigarette smell', 4);
```

6.8 Loyalty Cards Table

Create Table

```
CREATE TABLE loyalty_cards (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id bigint UNSIGNED NOT NULL,
  stamps int DEFAULT 0,
  unread_stamps int DEFAULT 0,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE
);
```

Insert 20 Values into Table

```
INSERT INTO loyalty_cards (user_id, stamps, unread_stamps)
VALUES
    (1, 5, 0),
    (2, 3, 1),
    (3, 8, 0),
    (4, 0, 0),
    (5, 2, 2),
    (1, 0, 0),
    (2, 1, 0),
    (3, 9, 0),
    (5, 4, 1),
    (6, 0, 0),
    (7, 2, 0),
    (8, 6, 0),
    (9, 1, 1),
    (10, 3, 0),
    (11, 7, 0),
    (12, 0, 0),
    (13, 2, 0),
    (14, 5, 0),
    (15, 1, 0),
    (16, 4, 2);
```

6.9 Maintenance Logs Table

Create Table

```
CREATE TABLE maintenance_logs (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  vehicle_id bigint UNSIGNED NOT NULL,
  user_id bigint UNSIGNED NOT NULL,
  date date NOT NULL,
  description varchar(255) NOT NULL,
  cost decimal(8,2) DEFAULT 0.00,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (vehicle_id) REFERENCES vehicles (id) ON DELETE CASCADE
);
```

Insert 20 Values into Table

```
INSERT INTO maintenance_logs (vehicle_id, user_id, date, description, cost)
VALUES
    (1, 4, '2025-09-01', 'Regular oil change and filter replacement', 120.00),
    (2, 4, '2025-09-05', 'Replaced 2 worn front tyres', 400.00),
    (3, 4, '2025-09-10', 'Brake pad inspection and cleaning', 80.00),
    (4, 4, '2025-09-12', 'Replaced dead battery', 250.00),
    (5, 4, '2025-09-15', 'Aircon gas refill and cleaning', 150.00),
    (1, 4, '2025-09-20', 'Replaced front wiper blades', 45.00),
    (2, 4, '2025-09-25', 'Puspakom routine inspection fee', 50.00),
   (3, 4, '2025-10-01', 'Repair dent on rear left door', 300.00),
   (4, 4, '2025-10-05', 'Major service (Spark plugs, ATF, Coolant)', 600.00),
   (5, 4, '2025-10-08', 'Headlight bulb replacement', 30.00)
   (1, 4, '2025-10-15', 'Wheel alignment and balancing', 60.00),
   (2, 4, '2025-10-20', 'Interior deep cleaning (vomit)', 100.00),
   (3, 4, '2025-10-22', 'Fixed loose side mirror', 50.00),
   (4, 4, '2025-11-01', 'Oil top-up', 20.00),
   (5, 4, '2025-11-05', 'Replaced cabin air filter', 40.00),
   (1, 4, '2025-11-10', 'Emergency jump start service', 80.00),
   (2, 4, '2025-11-15', 'Patch punctured tyre', 15.00),
   (3, 4, '2025-11-20', 'Gearbox seal replacement', 800.00),
   (4, 4, '2025-11-25', 'Polishing and waxing', 180.00),
   (5, 4, '2025-12-01', 'Routine oil change', 120.00);
```

6.10 Notifications Table

Create Table

```
CREATE TABLE notifications (
  id char(36) NOT NULL,
  type varchar(255) NOT NULL,
  notifiable_type varchar(255) NOT NULL,
  notifiable_id bigint UNSIGNED NOT NULL,
  data text NOT NULL,
  read_at timestamp NULL DEFAULT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert 20 Values into Table

```
INSERT INTO notifications (id, type, notifiable_type, notifiable_id, data,
read_at)
VALUES
    ('038c7126-af4c-427a-9951-bd54694ac559', 'App\\Notifications\\FineIssued',
'App\\Models\\User', 1, '{"booking_id": 51, "message": "You have been issued a
fine for Booking #51.", "amount": 50.00}', '2026-01-20 17:31:34'),
    ('471a986f-4467-4db2-82b1-1ada22862cf9',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 7,
'{"booking_id": 50, "status": "Approved", "message": "Your booking #50 has been
approved."}', '2026-01-20 21:38:35'),
     ('a5ae0d95-7700-4e92-a935-ab568d2eaf0b',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 1,
'{"booking_id": 51, "status": "Approved", "message": "Your booking #51 has been
approved."}', '2026-01-20 16:52:25'),
    ('c81e728d-9d4c-4712-b5e1-7d1a2c3f4e5a',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 2,
'{"booking_id": 52, "status": "Cancelled", "message": "Your booking #52 has been
cancelled."}', NULL),
    ('d92f839e-1e5d-4823-c6f2-8e2b3d4a5f6b',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 3,
'{"booking_id": 53, "status": "Completed", "message": "Your booking #53 has been
marked as completed. Thank you!"}', '2026-01-21 09:15:00'),
    ('e13a940f-2f6e-5934-d7g3-9f3c4e5b6g7c', 'App\\Notifications\\FineIssued',
'App\\Models\\User', 2, '{"booking_id": 54, "message": "You have been issued a
fine for Booking #54.", "amount": 30.00}', NULL),
    ('f24b051g-3g7f-6045-e8h4-0g4d5f6c7h8d',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 4,
'{"booking_id": 55, "status": "Pending", "message": "Your booking #55 is pending
approval."}', NULL),
    ('g35c162h-4h8g-7156-f9i5-1h5e6g7d8i9e',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 5,
'{"booking_id": 56, "status": "Rejected", "message": "Your booking #56 has been
rejected due to unavailable vehicle."}', '2026-01-22 14:20:00'),
```

```
    ('h46d273i-5i9h-8267-g0j6-2i6f7h8e9j0f', 'App\\Notifications\\FinePaid',
'App\\Models\\User', 1, '{"booking_id": 51, "message": "Thank you. Your fine for
Booking #51 has been paid."}', '2026-01-23 10:00:00'),
    ('i57e384j-6j0i-9378-h1k7-3j7g8i9f0k1g',
'App\\Notifications\\InspectionCompleted', 'App\\Models\\User', 6, '{"booking_id":
57, "message": "Vehicle inspection for booking #57 is complete."}', NULL),
    ('j68f495k-7k1j-0489-i2l8-4k8h9j0g1l2h',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 7,
'{"booking_id": 58, "status": "Confirmed", "message": "Your booking #58 is
confirmed."}', NULL),
    ('k79g506l-8l2k-1590-j3m9-5l9i0k1h2m3i',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 1,
'{"booking_id": 59, "status": "Returning", "message": "Please return vehicle for
booking #59 by 5PM."}', NULL),
    ('l80h617m-9m3l-2601-k4n0-6m0j1l2i3n4j', 'App\\Notifications\\FineIssued',
'App\\Models\\User', 3, '{"booking_id": 60, "message": "Late return fine issued
for Booking #60.", "amount": 15.00}', NULL),
    ('m91i728n-0n4m-3712-l5o1-7n1k2m3j4o5k',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 2,
'{"booking_id": 61, "status": "Approved", "message": "Booking #61 Approved."}',
'2026-01-25 08:30:00'),
    ('n02j839o-1o5n-4823-m6p2-8o2l3n4k5p6l',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 4,
'{"booking_id": 62, "status": "Completed", "message": "Booking #62 Completed."}',
'2026-01-26 12:00:00'),
    ('o13k940p-2p6o-5934-n7q3-9p3m4o5l6q7m',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 5,
'{"booking_id": 63, "status": "Pending", "message": "Booking #63 Verification
needed."}', NULL),
    ('p24l051q-3q7p-6045-o8r4-0q4n5p6m7r8n', 'App\\Notifications\\FineIssued',
'App\\Models\\User', 6, '{"booking_id": 64, "message": "Damage fine issued for
Booking #64.", "amount": 150.00}', NULL),
    ('q35m162r-4r8q-7156-p9s5-1r5o6q7n8s9o',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 7,
'{"booking_id": 65, "status": "Approved", "message": "Booking #65 Approved."}',
'2026-01-28 15:45:00'),
    ('r46n273s-5s9r-8267-q0t6-2s6p7r8o9t0p',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 8,
'{"booking_id": 66, "status": "Cancelled", "message": "Booking #66 Cancelled."}',
'2026-01-29 09:10:00'),
    ('s57o384t-6t0s-9378-r1u7-3t7q8s9p0u1q',
'App\\Notifications\\BookingStatusUpdated', 'App\\Models\\User', 9,
'{"booking_id": 67, "status": "Approved", "message": "Booking #67 Approved."}',
NULL);
```

6.11 Password Reset Tokens Table

Create Table

```
CREATE TABLE password_reset_tokens (
  email varchar(60) NOT NULL,
  token varchar(255) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (email)
);
```

Insert 20 Values into Table

```
INSERT INTO password_reset_tokens (email, token, created_at)
VALUES
    ('ali@graduate.utm.my', '$2y$10$token_hash_string_01', NOW()),
    ('siti@graduate.utm.my', '$2y$10$token_hash_string_02', NOW()),
    ('chong@graduate.utm.my', '$2y$10$token_hash_string_03', NOW()),
    ('muthu@graduate.utm.my', '$2y$10$token_hash_string_04', DATE_SUB(NOW(),
INTERVAL 10 MINUTE)),
    ('admin@utm.my', '$2y$10$token_hash_string_05', DATE_SUB(NOW(), INTERVAL 30
MINUTE)),
    ('siti@graduate.utm.my', '$2y$10$token_hash_string_06', DATE_SUB(NOW(),
INTERVAL 2 HOUR)),
    ('aiysah@graduate.utm.my', '$2y$10$token_hash_string_07', DATE_SUB(NOW(),
INTERVAL 5 HOUR)),
    ('muthu@graduate.utm.my', '$2y$10$token_hash_string_08', DATE_SUB(NOW(),
INTERVAL 1 DAY)),
    ('haowei@graduate.utm.my', '$2y$10$token_hash_string_09', '2025-10-01
10:00:00'),
    ('jingning@graduate.utm.my', '$2y$10$token_hash_string_10', '2025-10-05
14:30:00'),
    ('daniel@graduate.utm.my', '$2y$10$token_hash_string_11', DATE_SUB(NOW(),
INTERVAL 15 MINUTE)),
    ('adam@graduate.utm.my', '$2y$10$token_hash_string_12', DATE_SUB(NOW(),
INTERVAL 45 MINUTE)),
    ('sivaraj@graduate.utm.my', '$2y$10$token_hash_string_13', DATE_SUB(NOW(),
INTERVAL 3 HOUR)),
    ('wiliam@graduate.utm.my', '$2y$10$token_hash_string_14', DATE_SUB(NOW(),
INTERVAL 12 HOUR)),
    ('choonkeat@graduate.utm.my', '$2y$10$token_hash_string_15', DATE_SUB(NOW(),
INTERVAL 2 DAY)),
    ('aiman@graduate.utm.my', '$2y$10$token_hash_string_16', DATE_SUB(NOW(),
INTERVAL 1 WEEK)),
    ('iman@graduate.utm.my', '$2y$10$token_hash_string_17', NOW()),
    ('othman@graduate.utm.my', '$2y$10$token_hash_string_18', NOW()),
    ('zari@graduate.utm.my', '$2y$10$token_hash_string_19', DATE_SUB(NOW(),
INTERVAL 5 MINUTE)),
    ('imran@graduate.utm.my', '$2y$10$token_hash_string_20', DATE_SUB(NOW(),
INTERVAL 20 MINUTE));
```

6.12 Payments Table

Create Table

```
CREATE TABLE payments (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  booking_id bigint UNSIGNED NOT NULL,
  amount decimal(8,2) NOT NULL,
  payment_method varchar(50) DEFAULT 'QR',
  receipt_image varchar(255) NULL,
  status varchar(30) DEFAULT 'Pending',
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (booking_id) REFERENCES bookings (id) ON DELETE CASCADE
);
```

Insert 20 Values into Table

```
INSERT INTO payments (booking_id, amount, payment_method, receipt_image, status)
VALUES
    (1, 32.00, 'Direct Transfer', 'receipts/pay_01.jpg', 'Approved'),
    (2, 108.00, 'QR', 'receipts/pay_02.jpg', 'Approved'),
    (3, 32.00, 'QR', 'receipts/pay_03.jpg', 'Pending'),
    (4, 120.00, 'Direct Transfer', 'receipts/pay_04.jpg', 'Approved'),
    (6, 60.00, 'Direct Transfer', 'receipts/pay_05.jpg', 'Refunded'),
    (5, 432.00, 'QR', 'receipts/pay_06.jpg', 'Approved'),
    (7, 36.00, 'QR', 'receipts/pay_07_blur.jpg', 'Rejected'),
    (2, 15.00, 'QR', 'receipts/pay_08.jpg', 'Approved'),
    (5, 200.00, 'Direct Transfer', 'receipts/dep_05.jpg', 'Approved'),
    (8, 8.00, 'Direct Transfer', NULL, 'Failed'),
    (9, 240.00, 'QR', 'receipts/pay_11.jpg', 'Approved'),
    (10, 90.00, 'Direct Transfer', 'receipts/pay_12.jpg', 'Approved'),
    (11, 60.00, 'QR', 'receipts/pay_13.jpg', 'Pending'),
    (12, 16.00, 'QR', 'receipts/pay_14.jpg', 'Approved'),
    (13, 108.00, 'Direct Transfer', 'receipts/pay_15.jpg', 'Refunded'),
    (14, 40.00, 'QR', 'receipts/pay_16.jpg', 'Approved'),
    (15, 216.00, 'Direct Transfer', 'receipts/pay_17.jpg', 'Approved'),
    (16, 48.00, 'QR', 'receipts/pay_18.jpg', 'Pending'),
    (17, 16.00, 'QR', 'receipts/pay_19.jpg', 'Approved'),
    (18, 60.00, 'QR', 'receipts/pay_20.jpg', 'Rejected');
```

6.13 Pricing Rules Table

Create Table

```
CREATE TABLE pricing_rule_rates (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  pricing_tier_id bigint(20) UNSIGNED NOT NULL,
  hour_limit int NOT NULL,
  price decimal(8,2) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (pricing_tier_id) REFERENCES pricing_tiers (id),
);
```

Insert Default Values into Table

```
INSERT INTO pricing_rule_rates (pricing_tier_id, hour_limit, price)
VALUES
    (1, 1, 35.00),
    (1, 3, 55.00),
    (1, 5, 65.00),
    (1, 7, 70.00),
    (1, 9, 85.00),
    (1, 12, 95.00),
    (1, 24, 120.00),
    (2, 1, 40.00),
    (2, 3, 60.00),
    (2, 5, 70.00),
    (2, 7, 75.00),
    (2, 9, 90.00),
    (2, 12, 115.00),
    (2, 24, 140.00),
    (3, 1, 40.00),
    (3, 3, 60.00),
    (3, 5, 70.00),
    (3, 7, 75.00),
    (3, 9, 90.00),
    (3, 12, 100.00),
    (3, 24, 130.00),
    (4, 1, 50.00),
    (4, 3, 70.00),
    (4, 5, 80.00),
    (4, 7, 85.00),
    (4, 9, 95.00),
    (4, 12, 125.00),
    (4, 24, 150.00),
    (5, 1, 10.00),
    (5, 3, 16.00),
    (5, 5, 25.00),
    (5, 7, 31.00),
```

```
(5, 9, 36.00),
(5, 12, 40.00),
(5, 24, 43.00),
(6, 1, 9.00),
(6, 3, 17.00),
(6, 5, 24.00),
(6, 7, 30.00),
(6, 9, 35.00),
(6, 12, 39.00),
(6, 24, 42.00);
```

6.14 Pricing Tiers Table

Create Table

```
CREATE TABLE pricing_tiers (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  name varchar(50) NOT NULL,
  description text NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert Default Values into Table

```
INSERT INTO pricing_tiers (name, description)
VALUES
    ('Axia Tier 1', 'Standard pricing for Perodua Axia'),
    ('Axia Tier 2', 'Premium pricing for Perodua Axia (New Facelift)'),
    ('Myvi Tier', 'Standard pricing for Perodua Myvi'),
    ('Bezza Tier', 'Standard pricing for Perodua Bezza'),
    ('V010 Tier', 'Tier code V010'),
    ('V011 Tier', 'Tier code V011');
```

6.15 Promotions Table

Create Table

```
CREATE TABLE promotions (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  code varchar(30) NOT NULL,
  description varchar(255),
  discount_value decimal(8,2) NOT NULL,
  type varchar(20) NOT NULL,
  start_date date NOT NULL,
  end_date date NOT NULL,
  is_active tinyint(1) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert Default Values into Table

```
INSERT INTO promotions (code, description, discount_value, type, start_date,
end_date, is_active)
VALUES
('REWARD_3_STAMPS', '10% Off (3 Stamps)', 10.00, 'percentage', NULL, NULL, 1),
('REWARD_6_STAMPS', '15% Off (6 Stamps)', 15.00, 'percentage', NULL, NULL, 1),
('REWARD_9_STAMPS', '20% Off (9 Stamps)', 20.00, 'percentage', NULL, NULL, 1),
('REWARD_12_STAMPS', '25% Off (12 Stamps)', 25.00, 'percentage', NULL, NULL, 1),
('REWARD_15_STAMPS', '12 Hours Free (15 Stamps)', 12.00, 'free_hours', NULL, NULL,
1);
```

6.16 Users Table

Create Table

```
CREATE TABLE users (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  name varchar(100) NOT NULL,
  email varchar(60) NOT NULL UNIQUE,
  email_verified_at timestamp,
  password varchar(255) NOT NULL,
  role varchar(15) NOT NULL,
  matric_staff_id varchar(15) NOT NULL UNIQUE,
  matric_card_path varchar(255) NOT NULL,
  nric_passport_path varchar(255) NOT NULL,
  nric_passport varchar(255) NOT NULL,
  expired_date date NOT NULL,
  driving_license_path` varchar(255) NOT NULL,
  phone_number varchar(20) NOT NULL,
  address text NOT NULL,
  nationality varchar(100) NOT NULL,
  emergency_name varchar(100) NULL,
  emergency_contact varchar(20) NULL,
  emergency_relationship varchar(20) NULL,
  is_blacklisted tinyint(1) DEFAULT 0,
  blacklist_reason text,
  college_id bigint UNSIGNED NOT NULL,
  faculty_id bigint UNSIGNED NOT NULL,
  remember_token varchar(100),
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  bank_name varchar(255) NOT NULL,
  account_name varchar(255) NOT NULL,
  account_holder varchar(255) NOT NULL,
  salary varchar(10, 2) NULL,
  PRIMARY KEY (id)
  FOREIGN KEY (college_id) REFERENCES colleges (id),
  FOREIGN KEY (faculty_id) REFERENCES faculties (id),
);
```

Insert 20 Values into Table

```
INSERT INTO users (name, email, password, role, matric_staff_id, matric_card_path,
nric_passport_path, nric_passport, driving_license_path, expired_date,
phone_number, address, nationality, emergency_name, emergency_contact,
emergency_relationship,
college_id, faculty_id, bank_name, account_name, account_holder, salary)
VALUES
    ('Ali Bin Abu', 'ali@graduate.utm.my', '$2y$10$HaShEdPaSsWoRd', 'customer',
'A19EC0001', 'matric/ali.jpg', 'nric/ali.jpg', '980101-01-1234',
'license/ali.jpg', '2028-05-20', '0123456789', 'Kolej Tun Dr Ismail, UTM',
'Malaysian',  'Abu Bakar', '0198765432', 'Father',
```

```
 1, 9, 'Maybank', '1122334455', 'Ali Bin Abu', NULL),
    ('John Smith', 'john@graduate.utm.my', '$2y$10$HaShEdPaSsWoRd', 'customer',
'A20EC0055', 'matric/john.jpg', 'passport/john.jpg', 'P88997766',
'license/john.jpg', '2027-11-15','0189998888', 'Kolej 9, UTM', 'British',  'Mary
Smith', '+4477009000', 'Mother',
 12, 9, 'CIMB', '7055443322', 'John Smith', NULL),
    ('Encik Razak', 'razak@utm.my', '$2y$10$HaShEdPaSsWoRd', 'admin', 'S1234',
'staff/razak.jpg', 'nric/razak.jpg', '850505-01-5555', 'license/razak.jpg',
'2030-01-01',
 '0135554444', 'Taman Universiti, Skudai', 'Malaysian',  'Aminah', '0132221111',
'Spouse',
 20, 10, 'Bank Islam', '1209876543', 'Razak Bin Ahmad', '4500.00'),
    ('Siti Sarah', 'siti@graduate.utm.my', '$2y$10$HaShEdPaSsWoRd', 'customer',
'A19EC0002',  'matric/siti.jpg', 'nric/siti.jpg', '990202-02-2345',
'license/siti.jpg', '2026-08-10',
 '0134567890', 'Kolej Tun Fatimah, UTM', 'Malaysian', 'Fatimah', '0187654321',
'Mother',
11, 7, 'CIMB', '2233445566', 'Siti Sarah', NULL),
    ('Chong Wei', 'chong@graduate.utm.my', '$2y$10$HaShEdPaSsWoRd', 'customer',
'A19EC0003',  'matric/chong.jpg', 'nric/chong.jpg', '000303-03-3456',
'license/chong.jpg', '2029-03-15','0145678901', 'Kolej Rahman Putra, UTM',
'Malaysian', 'Mr Chong', '0176543210', 'Father', 10, 5, 'Public Bank',
'3344556677', 'Chong Wei', NULL),
    ('Muthu Sami', 'muthu@graduate.utm.my', '$2y$10$HaShEdPaSsWoRd', 'customer',
'A19EC0004',  'matric/muthu.jpg', 'nric/muthu.jpg', '010404-04-4567',
'license/muthu.jpg', '2027-02-28', '0167890123', 'Kolej Daton Onn Jaafar, UTM',
'Malaysian',  'Sami', '0154321098', 'Father',  5, 4, 'RHB', '4455667788', 'Muthu
Sami', NULL),
    ('Puan Nor', 'nor@utm.my', '$2y$10$HaShEdPaSsWoRd', 'staff', 'S5678',
'staff/nor.jpg', 'nric/nor.jpg', '800606-06-6666', 'license/nor.jpg',
'2026-12-31', '0191112222', 'Taman Pulai Utama', 'Malaysian',  'Azman',
'0193334444', 'Spouse',  20, 10, 'Maybank', '1122112211', 'Nor Binti Ali',
'3200.00'),
    ('Lim Ah Meng', 'lim@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A21EC4444', 'm/08.jpg', 'n/08.jpg', '020101-01-0808', 'l/08.jpg', '2028-01-01',
'0112233445', 'KTDI', 'Malaysian', 'Lim Seng Kuat', '0112223334', 'Father', 1, 9,
'Public Bank', '88877766', 'Lim Ah Meng', NULL),
    ('Nurul Huda', 'huda@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A21EC5555', 'm/09.jpg', 'n/09.jpg', '020202-02-0909', 'l/09.jpg', '2027-02-01',
'0123344556', 'KTHO', 'Malaysian', 'Nur Aminah', '0123334445', 'Mother', 4, 12,
'Bank Islam', '99988877', 'Nurul Huda', NULL),
    ('Ravi Kumar', 'ravi@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A21EC6666', 'm/10.jpg', 'n/10.jpg', '020303-03-1010', 'l/10.jpg', '2029-03-01',
'0134455667', 'KDOJ', 'Malaysian', 'Sivaraj', '0134445556', 'Father', 5, 5,
'CIMB', '77766655', 'Ravi Kumar', NULL),
    ('Tan Mei Ling', 'mei@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A21EC7777', 'm/11.jpg', 'n/11.jpg', '020404-04-1111', 'l/11.jpg', '2026-04-01',
'0145566778', 'KDSE', 'Malaysian', ‘Lee Jing Jing', '0145556667', 'Mother', 6, 9,
'Hong Leong', '66655544', 'Tan Mei Ling', NULL),
    ('Aiman Alif', 'aiman@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A22EC8888', 'm/12.jpg', 'n/12.jpg', '030505-05-1212', 'l/12.jpg', '2028-05-01',
```

```
'0156677889', 'KP', 'Malaysian', 'Ahmad Fakhrul', '0156667778', 'Father', 7, 3,
'Maybank', '55544433', ‘Aiman Alif', NULL),
    ('Siti Aisyah', 'siti@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A22EC9999', 'm/13.jpg', 'n/13.jpg', '030606-06-1313', 'l/13.jpg', '2027-06-01',
'0167788990', 'KTR', 'Malaysian', 'Nur Iman', '0167778889', 'Mother', 8, 2, 'RHB',
'44433322', 'Siti Aisyah', NULL),
    ('Tan Lay Kuan', 'tan@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A23EC0000', 'm/14.jpg', 'n/14.jpg', '040707-07-1414', 'l/14.jpg', '2029-07-01',
'0178899001', 'KIDI', 'Malaysian', 'Tan Lay Meng', '0178889990', 'Sibling', 9, 8,
'Bank Rakyat', '33322211', 'Tan Lay Kuan', NULL),
    ('Wong Hao Hao', 'haohao@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A23EC1111', 'm/15.jpg', 'n/15.jpg', '040808-08-1515', 'l/15.jpg', '2026-08-01',
'0189900112', 'KRP', 'Malaysian', 'Wong Ni Jing', '0189990001', 'Sibling', 10, 11,
'AmBank', '22211100', 'Wong Hao Hao', NULL),
    ('Raja Kumar', 'raja@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A23EC2222', 'm/16.jpg', 'n/16.jpg', '040909-09-1616', 'l/16.jpg', '2028-09-01',
'0190011223', 'KTF', 'Malaysian', 'Jivanesh', '0190001112', 'Relative', 11, 4,
'BSN', '11100099', 'Raja Kumar', NULL),
    ('Abdul Alif', 'u17@graduate.utm.my', '$2y$10$H@ash', 'customer', 'A24EC3333',
'm/17.jpg', 'n/17.jpg', '051010-10-1717', 'l/17.jpg', '2027-10-01', '0111122334',
'K9K10', 'Malaysian', 'Nur Natasha', '0111112223', 'Relative', 12, 13, 'OCBC',
'00099988', 'Abdul Alif', NULL),
    ('Zarina Aisyah', 'aisyah@utm.my', '$2y$10$H@ash', 'topmanegement', 'S9999',
'm/18.jpg', 'n/18.jpg', '828282-82-8282', 'l/18.jpg', '2030-01-01', '0122233445',
'Staff Housing', 'Malaysian', 'Abdul Ahmad', '0122223334', 'Spouse', 20, 10,
'Maybank', '12344321', 'Zarina Aisyah', '3800.00'),
    ('Lee Lay Jia', 'lee@utm.my', '$2y$10$H@ash', 'staff', 'S8888', 'm/19.jpg',
'n/19.jpg', '909090-90-9090', 'l/19.jpg', '2032-01-01', '0133344556', 'Staff
Housing', 'Malaysian', 'Tan Chen Yan', '0133334445', 'Spouse', 20, 9, 'CIMB',
'43211234', 'Lee Lay Jia', '3500.00'),
    ('Ahmed Ali', 'ahmed@graduate.utm.my', '$2y$10$H@ash', 'customer',
'A24EC4444', 'm/20.jpg', 'p/20.jpg', 'Z123456', 'l/20.jpg', '2026-12-01',
'0144455667', 'KDOJ', 'Yemeni', 'Father', '+967112233', 'Father', 5, 6, 'Maybank',
'98989898', 'Ahmed Ali', NULL);
```

6.17 User Vouchers Table

Create Table

```
CREATE TABLE user_vouchers (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id varchar(15) UNSIGNED NOT NULL,
  voucher_id bigint UNSIGNED NOT NULL,
  used_at timestamp NULL DEFAULT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id),
  FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE,
  FOREIGN KEY (voucher_id) REFERENCES vouchers (id) ON DELETE CASCADE
);
```

Insert 20 Values into Table

```
INSERT INTO user_vouchers (user_id, voucher_id, used_at)
VALUES
    ('A19EC0001', 1, '2025-10-01 10:00:00'),
    ('A19EC0002', 2, '2025-10-02 09:00:00'),
    ('A19EC0003', 1, '2025-10-05 12:00:00'),
    ('A19EC0001', 6, '2025-10-10 10:00:00'),
    ('A20EC0055', 6, '2025-11-01 08:00:00'),
    ('A19EC0004', 3, '2025-10-25 10:00:00'),
    ('A19EC0001', 2, NULL),
    ('A19EC0002', 3, NULL),
    ('A19EC0003', 4, NULL),
    ('A19EC0004', 1, NULL),
    ('S1234', 12, NULL),
    ('S5678', 12, NULL),
    ('A19EC0001', 7, '2025-08-31 10:00:00'),
    ('A19EC0002', 14, '2025-02-02 09:00:00'),
    ('A19EC0003', 11, '2025-06-15 12:00:00'),
    ('A21EC4444', 6, '2025-09-01 10:00:00'),
    ('A21EC5555', 6, '2025-09-02 11:00:00'),
    ('A21EC6666', 6, NULL),
    ('A21EC7777', 1, NULL),
    ('A19EC0001', 13, '2025-11-15 14:00:00');
```

6.18 Vehicles Table

Create Table

```
CREATE TABLE vehicles (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  vehicle_id_custom varchar(10) UNIQUE,
  plate_number varchar(10) NOT NULL UNIQUE,
  brand varchar(255) NOT NULL,
  model varchar(255) NOT NULL,
  year int NOT NULL,
  capacity int NOT NULL,
  price_per_hour decimal(8,2) NOT NULL,
  vehicle_image varchar(255) NULL,
  is_hasta_owned tinyint(1) DEFAULT 1,
  owner_name varchar(100),
  owner_ic_path varchar(255),
  owner_license_path varchar(255),
  geran_path varchar(255),
  insurance_cover_path varchar(255),
  current_fuel_bars int NOT NULL,
  status varchar(30) DEFAULT 'Available',
  type_id int,
  road_tax_expiry date NOT NULL,
  insurance_expiry date NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert Default Values into Table

```
INSERT INTO vehicles (vehicle_id_custom, plate_number, brand, model, year,
capacity, price_per_hour, vehicle_image, is_hasta_owned, owner_name,
owner_ic_path, owner_license_path, geran_path, insurance_cover_path,
current_fuel_bars, status, type_id, road_tax_expiry, insurance_expiry)
VALUES
    ('V001', 'MCP 6113', 'PERODUA', 'AXIA', 2014, 5, 10.00,
'PERODUA-Axia-5639_10.jpeg', 1, NULL, NULL, NULL, NULL, NULL, 8, 'Available', 1,
'2026-01-31', '2026-01-31'),
    ('V002', 'JQU 1957', 'PERODUA', 'AXIA', 2015, 5, 10.00,
'PERODUA-Axia-5639_10.jpeg', 1, NULL, NULL, NULL, NULL, NULL, 9, 'Unavailable', 1,
'2026-02-15', '2026-02-15'),
    ('V003', 'NDD 7803', 'PERODUA', 'AXIA', 2016, 5, 10.00,
'PERODUA-Axia-5639_10.jpeg', 1, NULL, NULL, NULL, NULL, NULL, 7, 'Available', 1,
'2026-03-20', '2026-03-20'),
    ('V004', 'CEX 5224', 'PERODUA', 'AXIA', 2024, 5, 10.00,
'PERODUA-Axia-5639_10.jpeg', 1, NULL, NULL, NULL, NULL, NULL, 9, 'Available', 1,
'2026-06-01', '2026-06-01'),
```

```
    ('V005', 'UTM 3365', 'PERODUA', 'AXIA', 2024, 5, 10.00,
'PERODUA-Axia-5639_10.jpeg', 1, NULL, NULL, NULL, NULL, NULL, 8, 'Available', 1,
'2026-06-15', '2026-06-15'),
    ('V006', 'JPN 1416', 'PERODUA', 'MYVI', 2013, 5, 10.00,
'perodua-myvi-color-797792.png', 1, NULL, NULL, NULL, NULL, NULL, 7, 'Available',
3, '2026-04-10', '2026-04-10'),
    ('V007', 'VC 6522', 'PERODUA', 'MYVI', 2016, 5, 10.00,
'perodua-myvi-color-797792.png', 1, NULL, NULL, NULL, NULL, NULL, 6, 'Available',
3, '2026-05-05', '2026-05-05'),
    ('V008', 'UTM 3855', 'PERODUA', 'BEZZA', 2023, 5, 10.00, 'bezza.jpg', 1, NULL,
NULL, NULL, NULL, NULL, 9, 'Available', 2, '2026-07-20', '2026-07-20'),
    ('V009', 'UTM 3857', 'PERODUA', 'BEZZA', 2023, 5, 10.00, 'bezza.jpg', 1, NULL,
NULL, NULL, NULL, NULL, 7, 'Available', 2, '2026-08-01', '2026-08-01'),
    ('V010', 'QRP 5205', 'HONDA', 'DASH 125', 2021, 2, 10.00, 'DASH125.png', 1,
NULL, NULL, NULL, NULL, NULL, 5, 'Available', 5, '2026-09-10', '2026-09-10'),
    ('V011', 'JWD 9496', 'HONDA', 'BEAT 110', 2023, 2, 10.00, 'Honda_Beat110.png',
1, NULL, NULL, NULL, NULL, NULL, 6, 'Available', 5, '2026-10-15', '2026-10-15');
```

6.19 Vehicle Types Table

Create Table

```
CREATE TABLE vehicle_types (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  code int NOT NULL,
  name varchar(30) NOT NULL,
  base_price decimal(8, 2) NOT NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  PRIMARY KEY (id)
);
```

Insert Default Values into Table

```
INSERT INTO vehicle_types (code, name, base_price)
VALUES
    (1, 'Compact (Axia)', 8.00),
    (2, 'Sedan (Bezza)', 10.00),
    (3, 'Hatchback (Myvi)', 9.00),
    (5, 'Motorcycle', 5.00);
```

6.20 Vouchers Table

Create Table

```
CREATE TABLE vouchers (
  id bigint UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id varchar(15) UNSIGNED NULL,
  code varchar(30) UNIQUE NOT NULL,
  name varchar(50) NOT NULL,
  type varchar(20) NOT NULL,
  value decimal(8, 2) NOT NULL,
  is_active tinyint(1) DEFAULT 1,
  expires_at timestamp NULL,
  created_at timestamp NULL DEFAULT NULL,
  updated_at timestamp NULL DEFAULT NULL,
  single_use tinyint(1),
  uses_remaining int NOT NULL,
  PRIMARY KEY (id)
  FOREIGN KEY (user_id) REFERENCES users (id) ON DELETE CASCADE,
);
```

Insert 20 Values in Table

```
INSERT INTO vouchers (user_id, code, name, type, value, is_active, expires_at,
single_use, uses_remaining)
VALUES
    (NULL, 'WELCOME2025', 'New Year Welcome', 'percentage', 10.00, 1, '2026-01-01
00:00:00', 0, 1000),
    ('A19EC0001', 'ALI-BDAY', 'Birthday Reward', 'fixed', 50.00, 1, '2026-06-30
00:00:00', 1, 1),
    (NULL, 'EXPIRED2024', 'Last Year Promo', 'percentage', 15.00, 0, '2024-12-31
23:59:59', 0, 0),
    (NULL, 'VIP50', 'VIP Discount', 'fixed', 50.00, 1, '2026-12-31 00:00:00', 0,
10),
    (NULL, 'STUDENT20', 'Student Discount', 'percentage', 20.00, 1, '2026-05-31
00:00:00', 0, 500),
    (NULL, 'FREE1HR', '1 Hour Free', 'free_hours', 1.00, 1, '2026-12-31 00:00:00',
1, 200),
    ('A19EC0002', 'SORRY10', 'Service Recovery', 'fixed', 10.00, 1, '2026-03-01
00:00:00', 1, 1),
    (NULL, 'WEEKEND15', 'Weekend Off', 'percentage', 15.00, 1, '2026-12-31
00:00:00', 0, 100),
    (NULL, 'STAFFPROMO', 'Staff Only', 'percentage', 25.00, 1, '2030-01-01
00:00:00', 0, 9999),
    (NULL, 'FLASH50', '50% Off Flash', 'percentage', 50.00, 1, '2025-02-05
23:59:59', 1, 5),
    (NULL, 'RAYA2025', 'Hari Raya Special', 'fixed', 30.00, 1, '2025-05-15
00:00:00', 0, 200),
    ('A19EC0003', 'CHONG-REW', 'Loyalty Reward', 'fixed', 20.00, 1, '2025-12-31
00:00:00', 1, 1),
```

```
    (NULL, 'MERDEKA68', 'Merdeka Promo', 'percentage', 31.00, 0, '2025-08-31
23:59:59', 0, 50),
    (NULL, 'CNY888', 'CNY Angpao', 'fixed', 8.88, 1, '2025-02-15 00:00:00', 0,
88),
    (NULL, 'DEEPAVALI', 'Festival Lights', 'percentage', 15.00, 1, '2025-11-01
00:00:00', 0, 100),
    (NULL, '1212SALE', '12.12 Mega Sale', 'percentage', 40.00, 0, '2025-12-12
23:59:59', 1, 50),
    (NULL, 'EARLYBIRD', 'Early Booking', 'percentage', 5.00, 1, '2026-12-31
00:00:00', 0, 500),
    ('S1234', 'ADMIN-TEST', 'System Test', 'fixed', 1.00, 1, '2030-12-31
00:00:00', 0, 999),
    (NULL, 'LOYAL10', 'Return Customer', 'percentage', 10.00, 1, '2026-12-31
00:00:00', 1, 300),
    (NULL, 'NEWUSER', 'First Time User', 'fixed', 15.00, 1, '2026-12-31 00:00:00',
1, 1000);
```

6.21 DML Skills

1. **SELECT with JOIN**
   Scenario: Retrieve full booking details including customer name, vehicle model, and total price for 'Approved' bookings.

```
SELECT
    b.id AS booking_id,
    u.name AS customer_name,
    v.brand,
    v.model,
    v.plate_number,
    b.start_date,
    b.end_date,
    b.total_price
FROM bookings b
JOIN users u ON b.user_id = u.id
JOIN vehicles v ON b.vehicle_id = v.id
WHERE b.status = 'Approved';
```

2. **UPDATE Statement**
   Scenario: Mark bookings as 'Completed' if their end_date has passed.

```
UPDATE bookings
SET status = 'Completed', updated_at = NOW()
WHERE status = 'Approved'
AND end_date < NOW();
```

3. **DELETE Statement**
   Scenario: Remove notifications older than 1 year that have already been read.

```
DELETE FROM notifications
WHERE read_at < DATE_SUB(NOW(), INTERVAL 1 YEAR)
AND read_at IS NOT NULL;
```

4. **Aggregate Function (SUM & COUNT)**
   Scenario: Calculate total revenue and number of completed bookings per vehicle, ordered by highest revenue.

```
SELECT
    v.plate_number,
    v.model,
    COUNT(b.id) AS total_bookings,
    SUM(b.total_price) AS total_revenue
FROM vehicles v
JOIN bookings b ON v.id = b.vehicle_id
WHERE b.status = 'Completed'
GROUP BY v.id
```

```
ORDER BY total_revenue DESC;
```

5. **Subquery with ORDER BY**
   Scenario: List details of "High Value Customers" who have made more than 3 bookings.

```
SELECT id, name, email, phone_number
FROM users
WHERE id IN (
    SELECT user_id
    FROM bookings
    GROUP BY user_id
    HAVING COUNT(id) > 3
)
ORDER BY name ASC;
```

7.0 Summary

This phase focuses on the logical database design and implementation for the Hasta Car Rental System. Based on the conceptual design produced in the previous phase, the conceptual ERD was transformed into a logical ERD by resolving complex relationships and deriving relational schemas suitable for implementation in a relational database system.

The normalization process was carried out from Unnormalized Form (UNF) up to Third Normal Form (3NF). All relations were further evaluated against Boyce-Codd Normal Form (BCNF), and no violations were identified where applicable.This normalization process helps eliminate data redundancy, avoid update anomalies, and ensure data integrity across the system. Each table was clearly defined with appropriate primary keys and foreign keys to maintain correct relationships between entities.

In addition, an updated data dictionary was produced to describe the structure and purpose of each table and attribute. SQL Data Definition Language (DDL) statements were used to create the database schema, while Data Manipulation Language (DML) statements were designed to support key system transactions such as booking creation, payment recording, vehicle updates, and report generation.

Overall, the logical database design successfully supports the system's transaction requirements and provides a structured and reliable foundation for system implementation. This database design ensures consistency, scalability, and efficient data management for the Hasta Car Rental System.

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